Name: BALA.S
Reg No: 192524427
Course & Code: CSA1522 Cloud Computing & Big Data

EXP NO 12: Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software, using virtual box.

Aim:

To create and configure a virtual machine with 1 CPU, 2 GB RAM, and 15 GB storage using VirtualBox as a Type-2 virtualization software.

Procedure:

Step 1: Open Oracle VirtualBox on the host computer.

Step 2: Select the already created Ubuntu VM and open Settings.

Step 3: Go to System → Motherboard and set the memory to 2048 MB (2 GB).

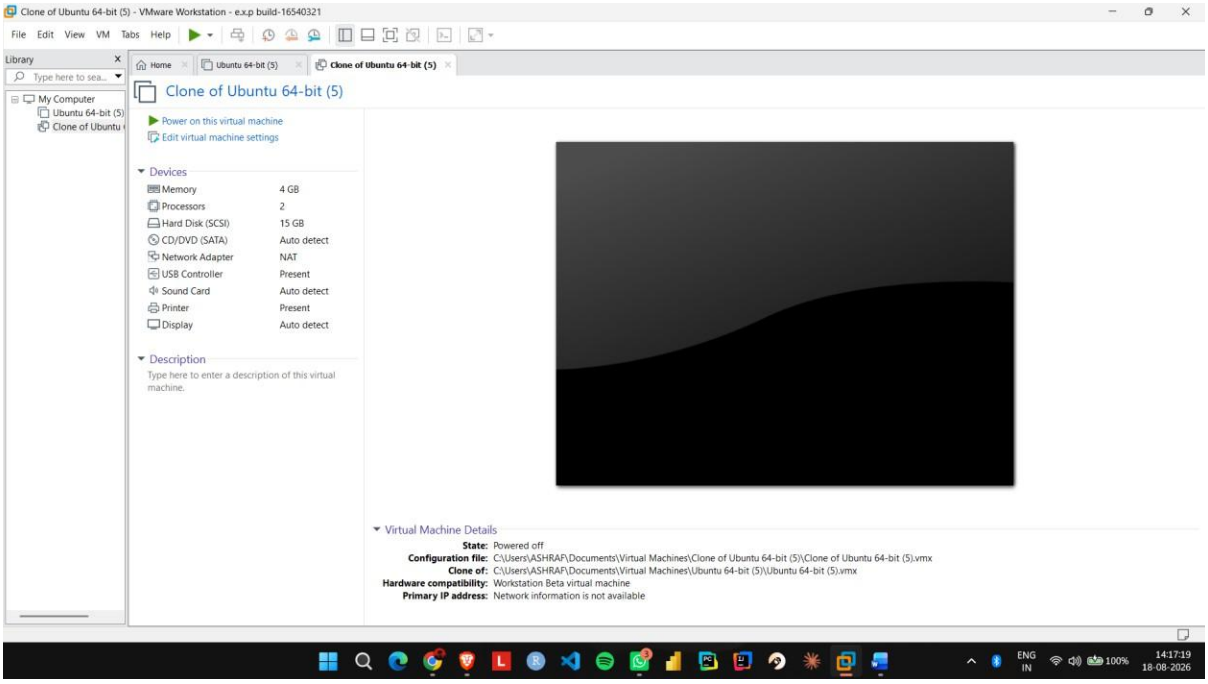
Step 4: Go to System → Processor and set the processor count to 1 CPU.

Step 5: Open Storage and verify that the virtual hard disk is configured with 15 GB storage.

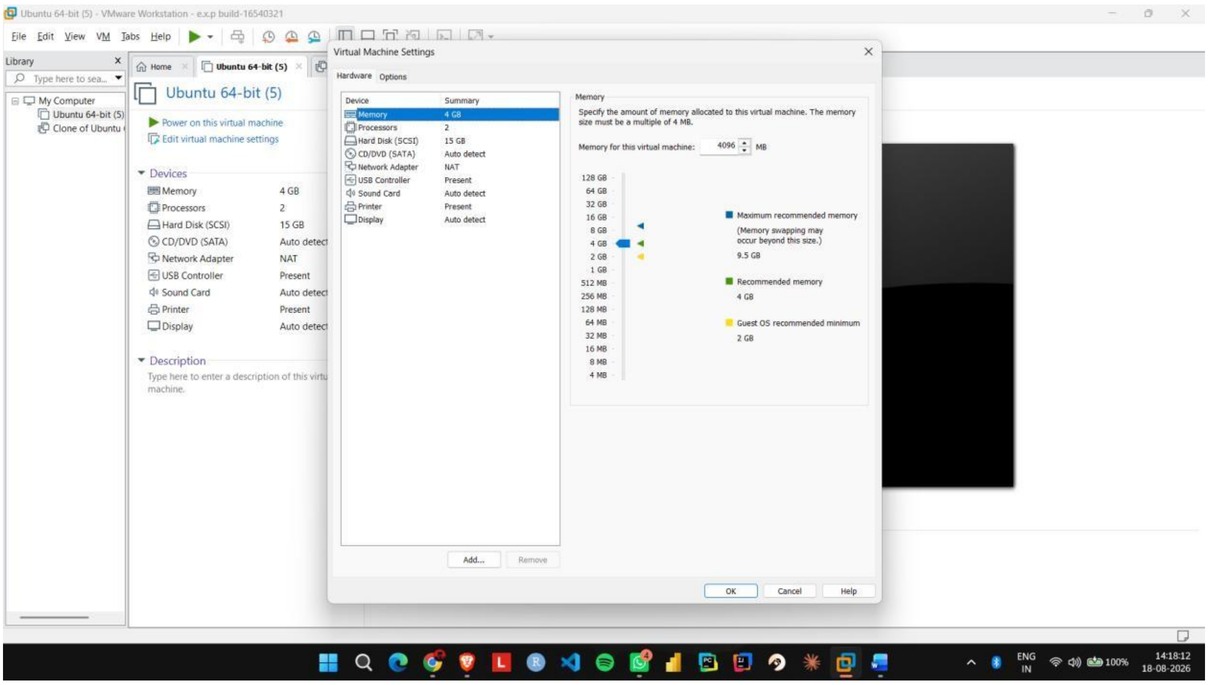
Step 6: Click OK, start the Ubuntu VM, and verify that it boots successfully.

IMPLEMENTATION:

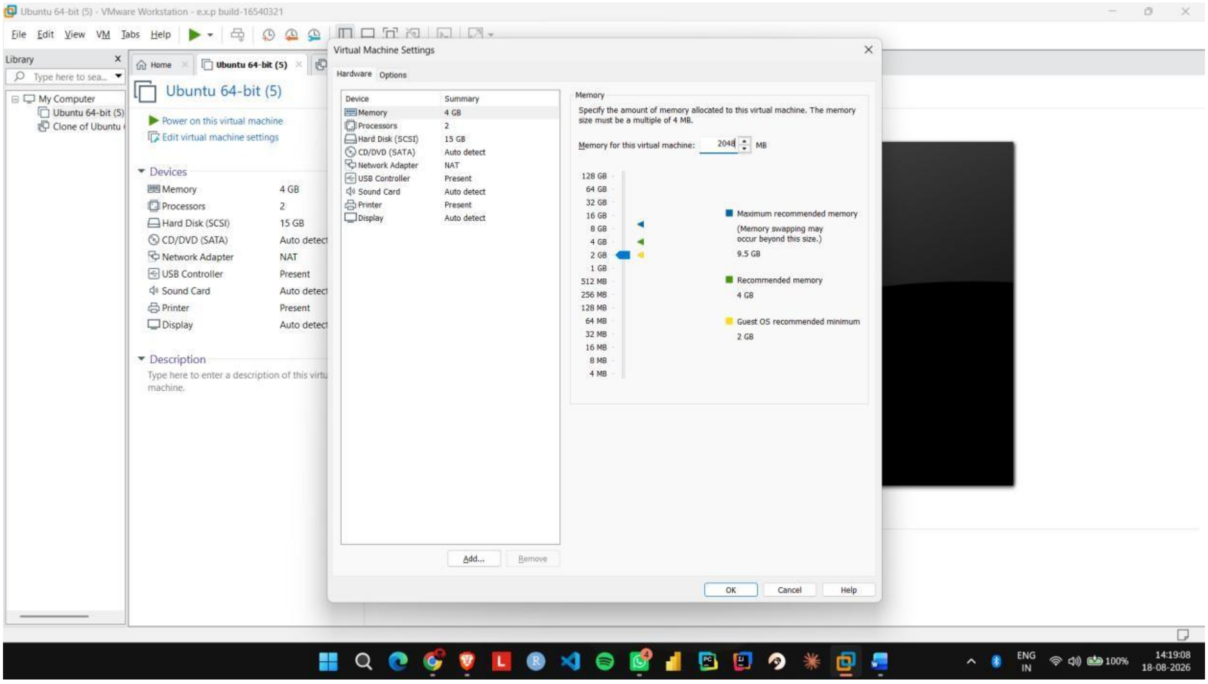
Step 1: Open Oracle VirtualBox on the host computer.



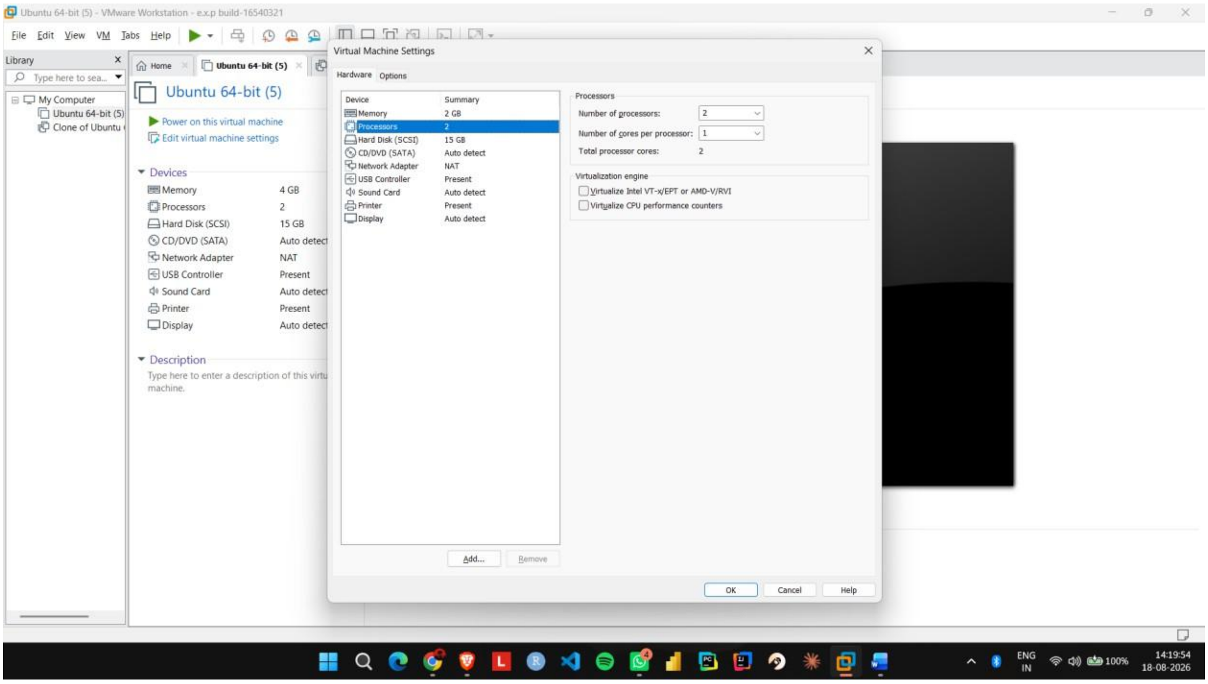
Step 2: Select the already created Ubuntu VM and open Settings.



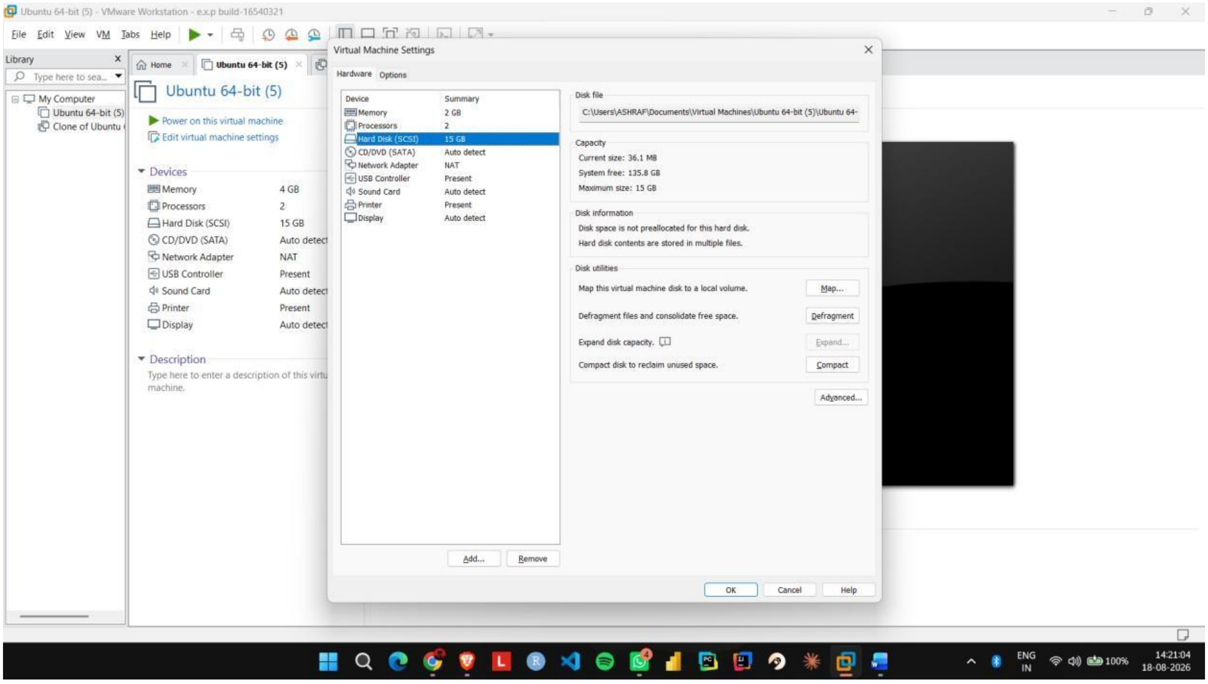
Step 3: Go to System → Motherboard and set the memory to 2048 MB (2 GB).



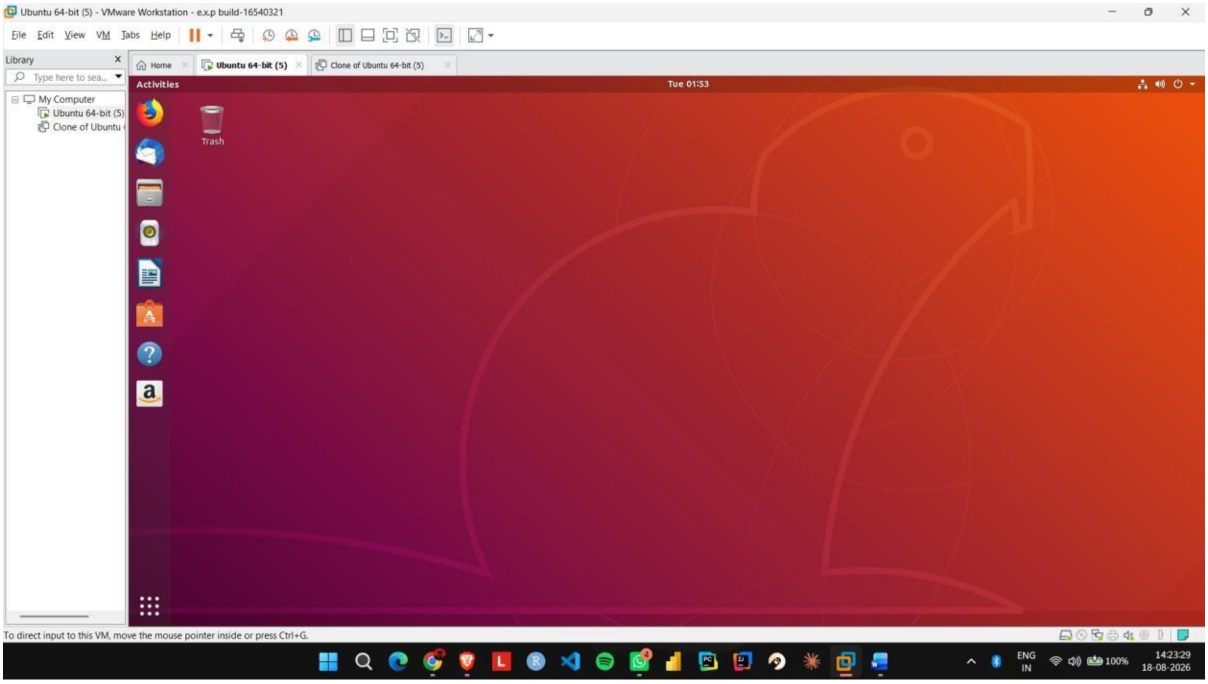
Step 4: Go to System → Processor and set the processor count to 1 CPU.



Step 5: Open Storage and verify that the virtual hard disk is configured with 15 GB storage.



Step 6: Click OK, start the Ubuntu VM, and verify that it boots successfully.



EXP NO 13: Create a Virtual Hard Disk and allocate the storage using virtual box.

Aim:

To create a virtual hard disk and allocate storage to it using VirtualBox.

Procedure:

Step 1: Open VMware Workstation and select the existing Ubuntu virtual machine.

Step 2: Click Edit virtual machine settings.

Step 3: Select Hard Disk and click Add to create a new virtual hard disk.

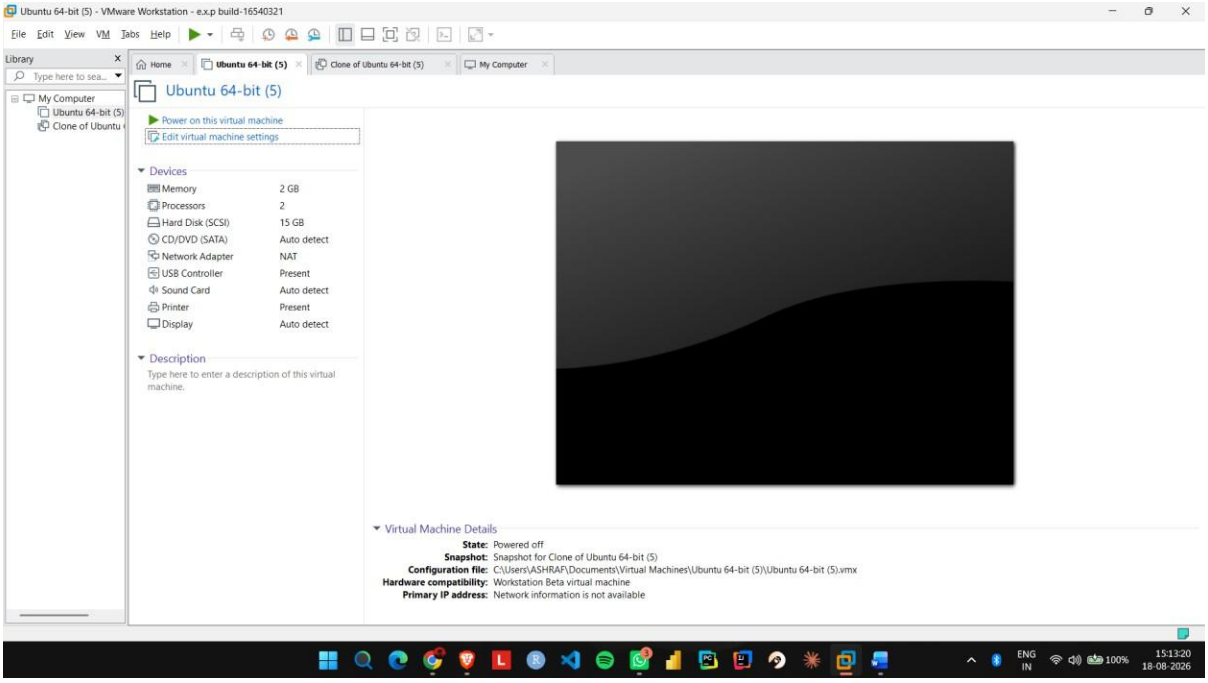
Step 4: Select Hard Disk → SCSI as the virtual disk type.

Step 5: Select Create a new virtual disk and allocate the required storage, such as 15 GB.

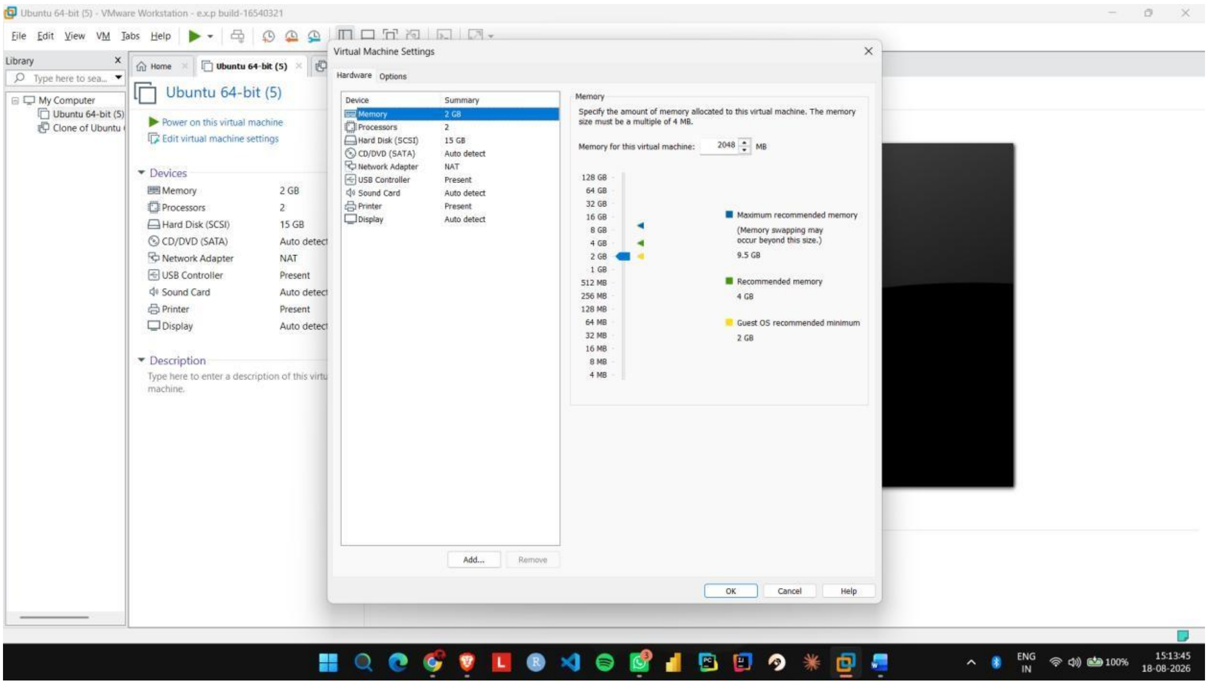
Step 6: Click Finish and verify that the new virtual hard disk is attached to the Ubuntu VM.

IMPLEMENTATION:

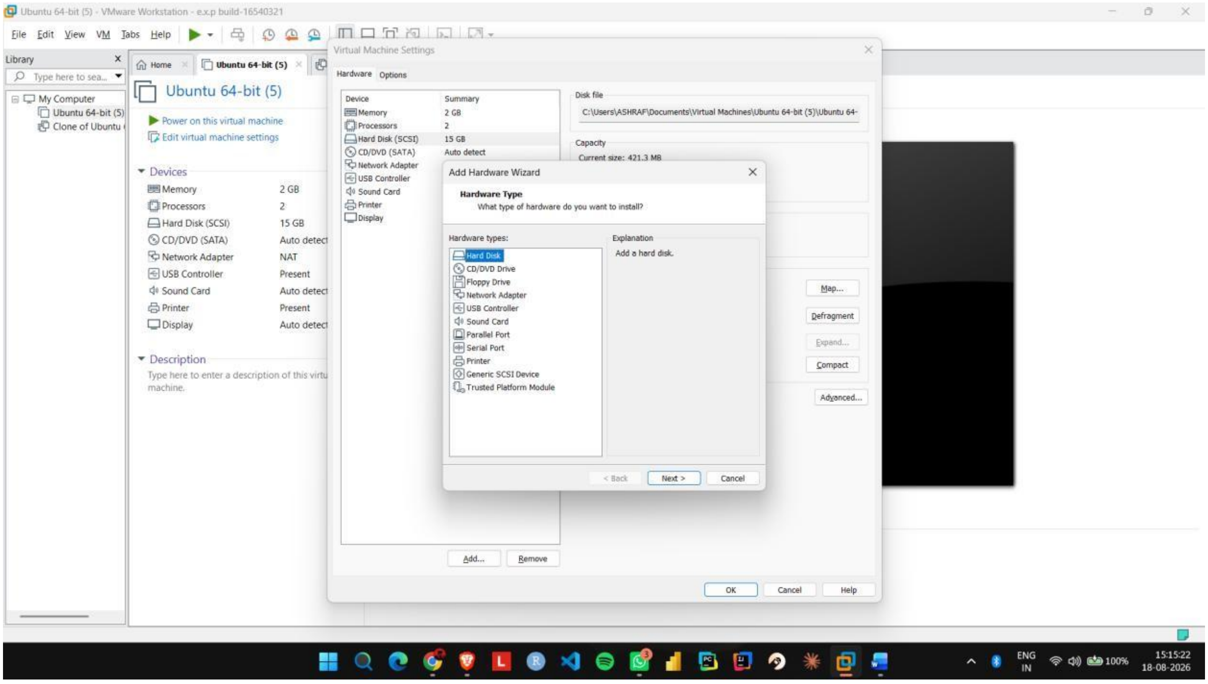
Step 1: Open VMware Workstation and select the existing Ubuntu virtual machine.



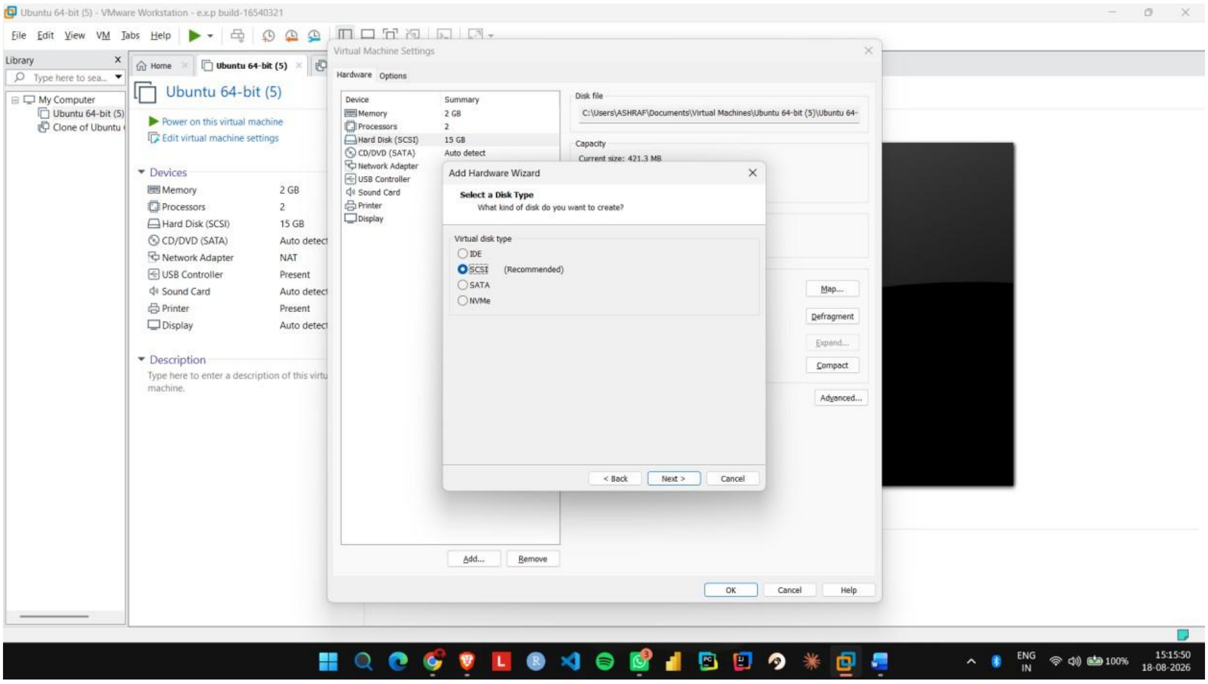
Step 2: Click Edit virtual machine settings.



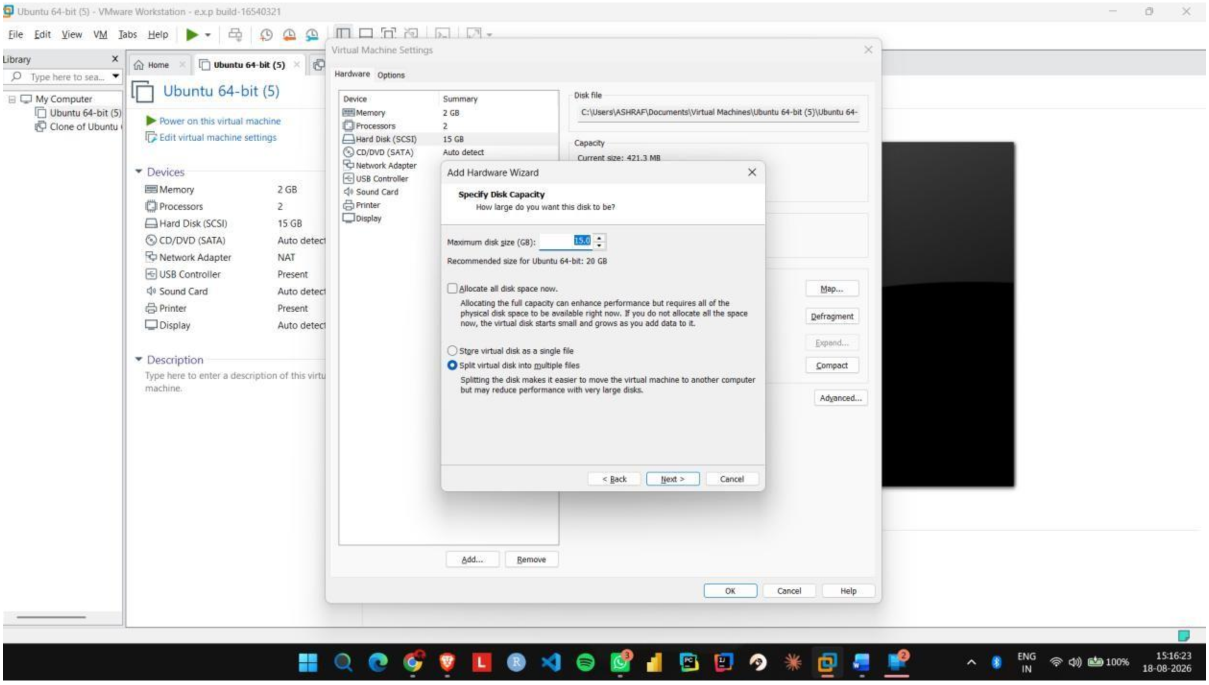
Step 3: Select Hard Disk and click Add to create a new virtual hard disk.



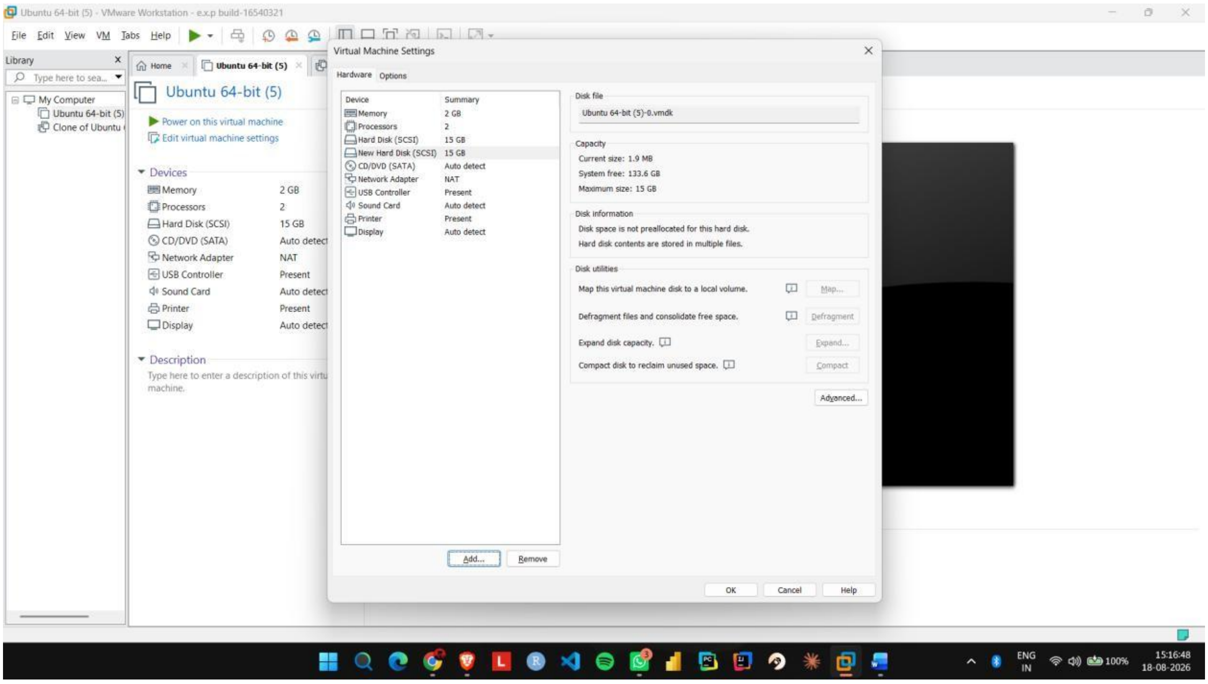
Step 4: Select Hard Disk → SCSI as the virtual disk type.



Step 5: Select Create a new virtual disk and allocate the required storage, such as 15 GB.



Step 6: Click Finish and verify that the new virtual hard disk is attached to the Ubuntu VM.



EXP NO 14: Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM using a virtual box.

Aim:

To create a snapshot of a virtual machine and restore the previous version using VMware Workstation.

Procedure:

Step 1: Start the Ubuntu VM in VMware Workstation and create a test file or folder.

Step 2: Shut down or suspend the VM and open the Snapshot option.

Step 3: Select Take Snapshot and enter a name such as Before_Changes.

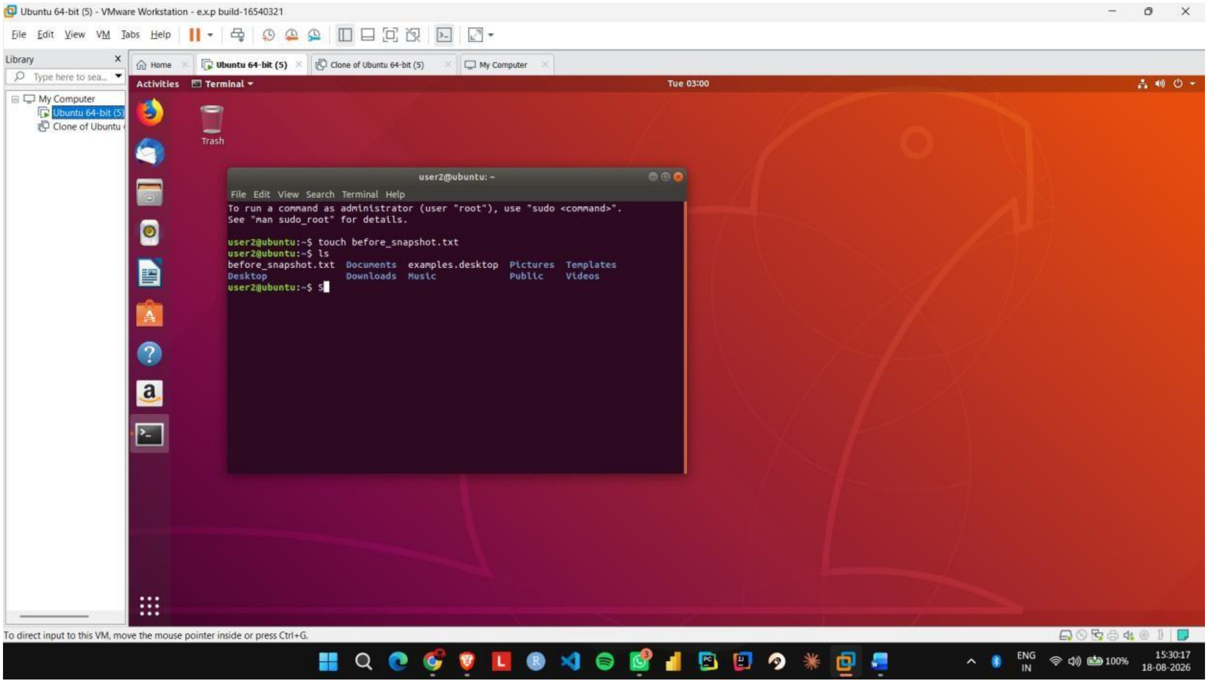
Step 4: Start the VM again and make some changes, such as creating a new file.

Step 5: Open Snapshot Manager, select the Before_Changes snapshot, and click Go To/Restore.

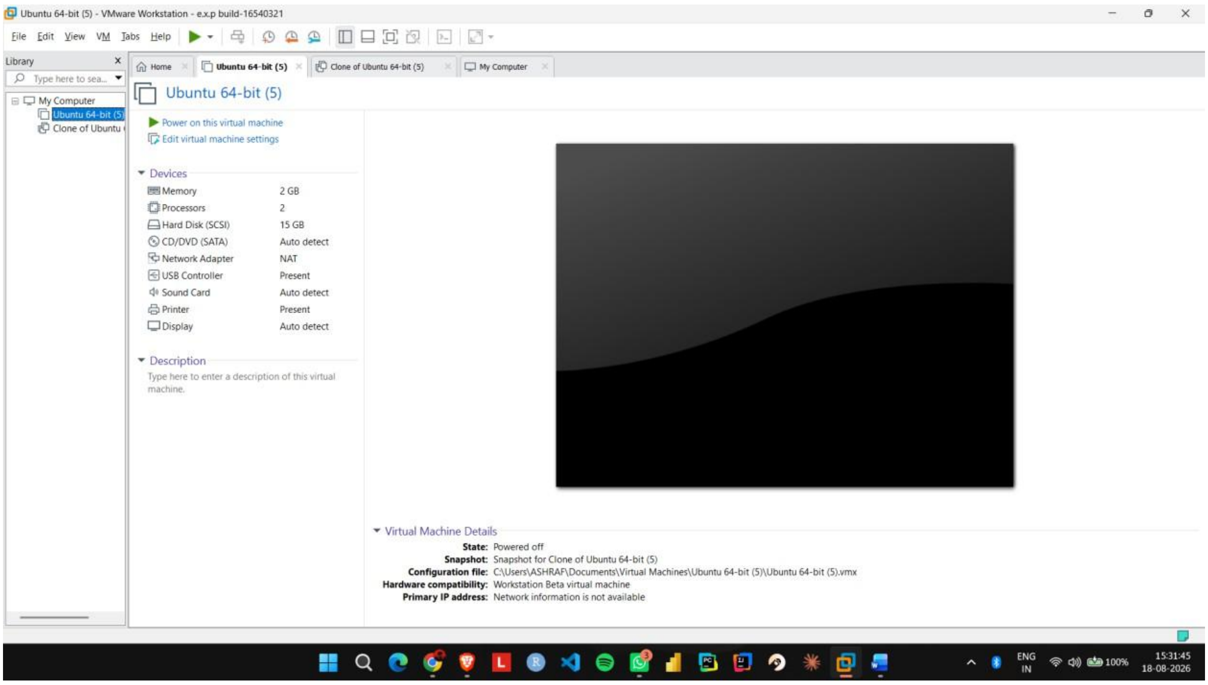
Step 6: Start the VM and verify that the changes made after the snapshot have been removed.

IMPLEMENTATION:

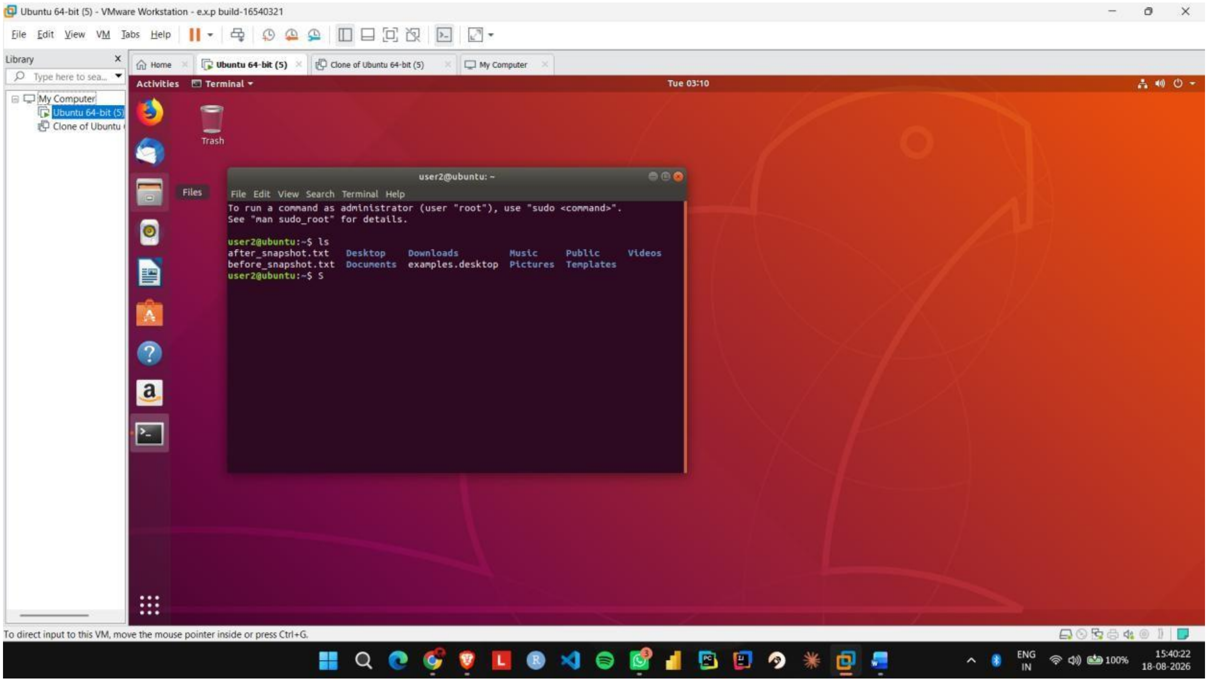
Step 1: Start the Ubuntu VM in VMware Workstation and create a test file or folder.



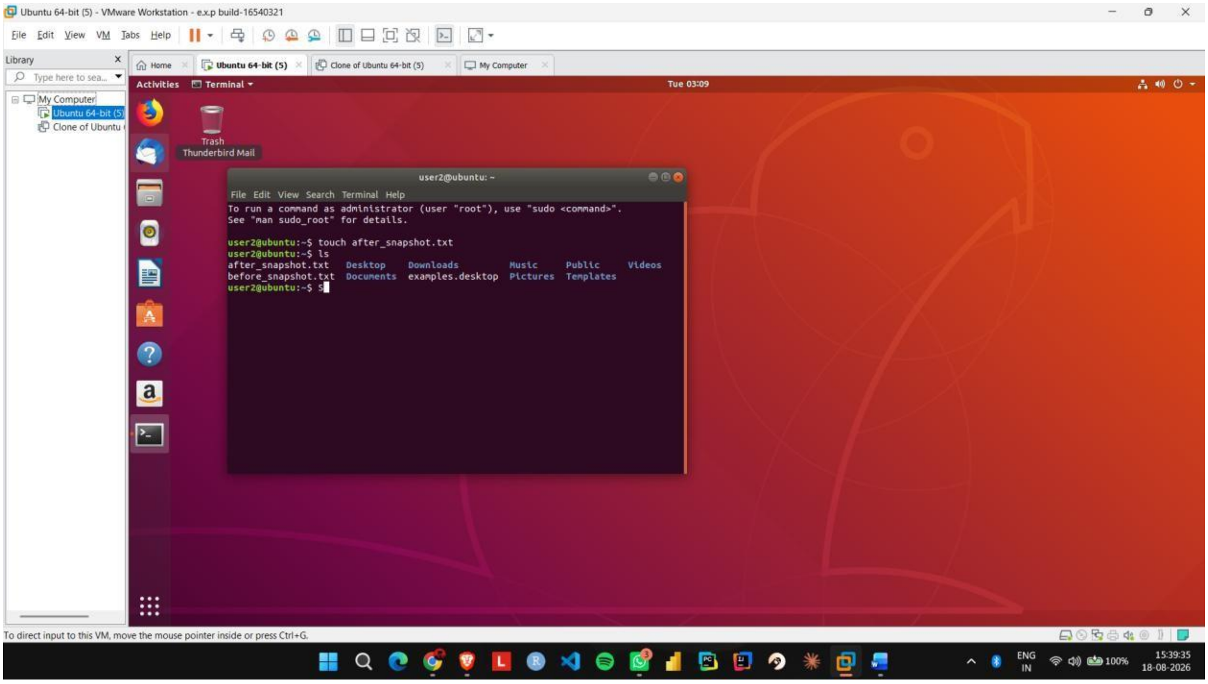
Step 2: Shut down or suspend the VM and open the Snapshot option.



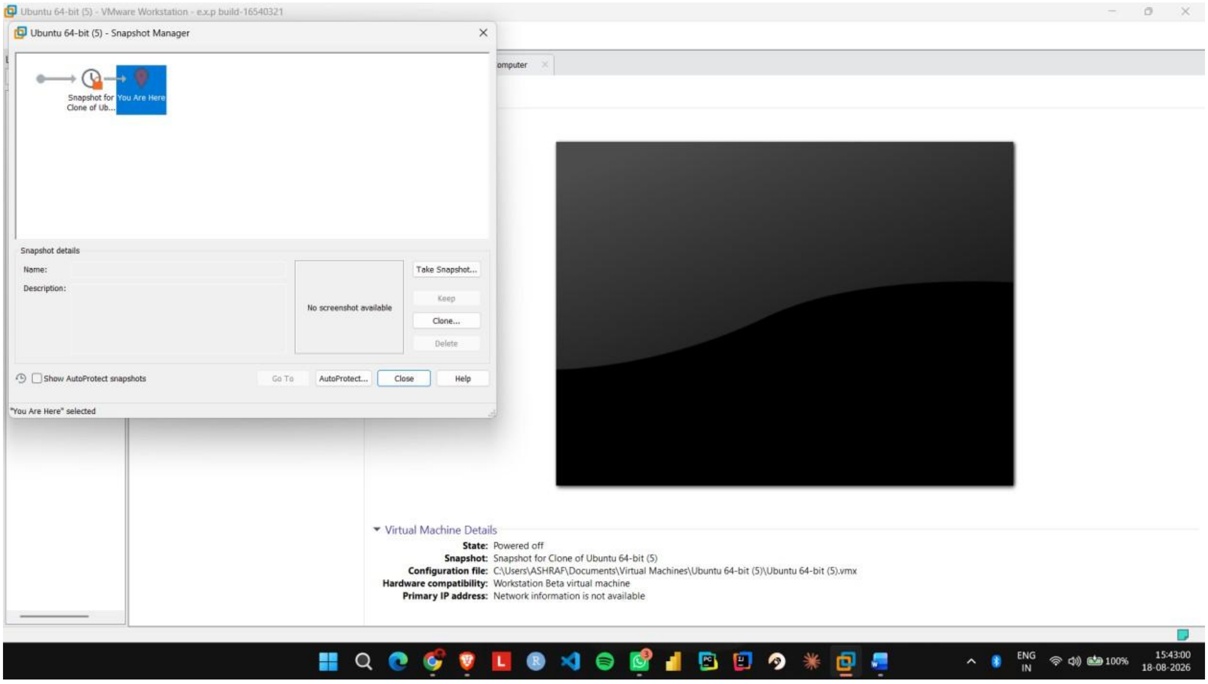
Step 3: Select Take Snapshot and enter a name such as Before_Changes.



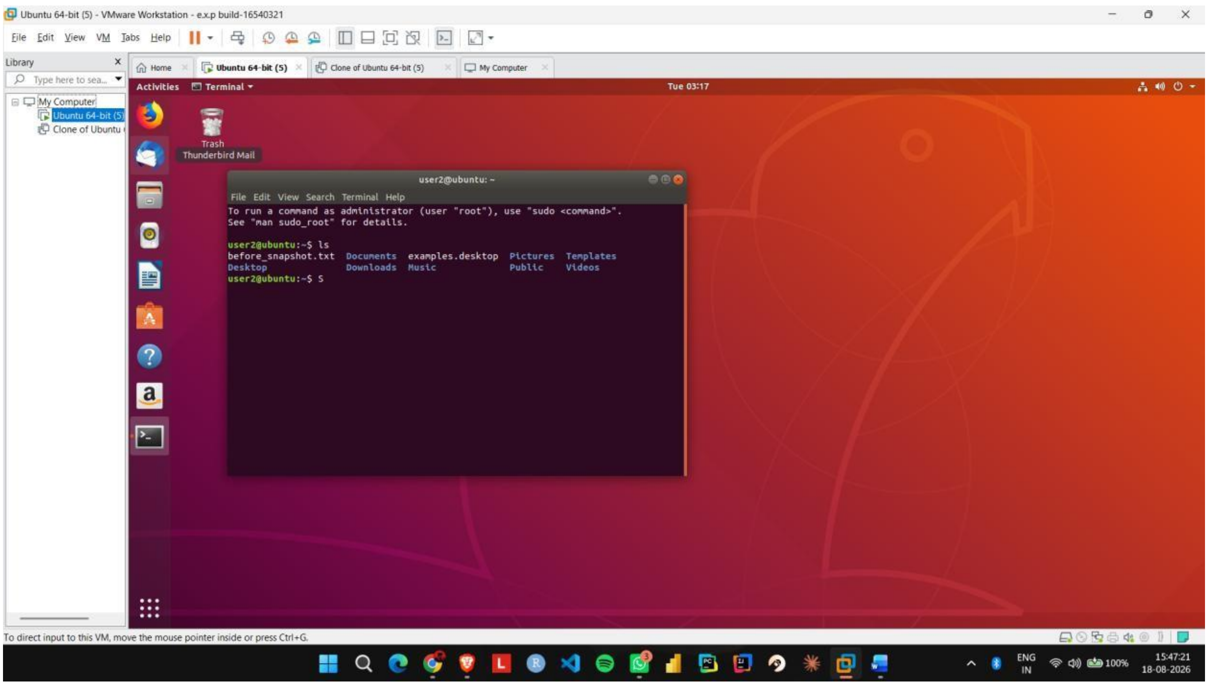
Step 4: Start the VM again and make some changes, such as creating a new file.



Step 5: Open Snapshot Manager, select the Before_Changes snapshot, and click Go To/Restore.



Step 6: Start the VM and verify that the changes made after the snapshot have been removed.



EXP NO 15: Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM using a virtual box.

Aim:

To create a clone of an existing virtual machine and test the cloned virtual machine using VMware Workstation.

Procedure:

Step 1: Shut down the existing Ubuntu VM completely.

Step 2: Open VMware Workstation and select VM → Manage → Clone.

Step 3: Select the current state of the virtual machine as the cloning source.

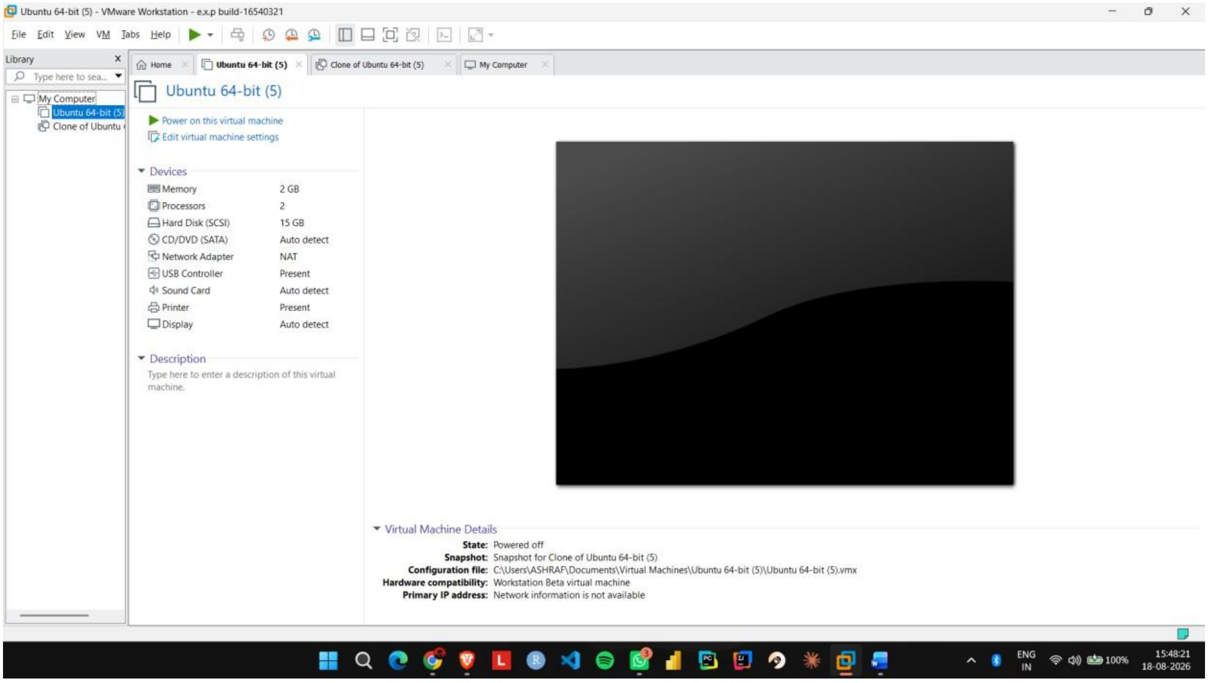
Step 4: Select Create a Full Clone and give the cloned VM a name such as Ubuntu_Clone.

Step 5: Complete the cloning process and start the Ubuntu_Clone virtual machine.

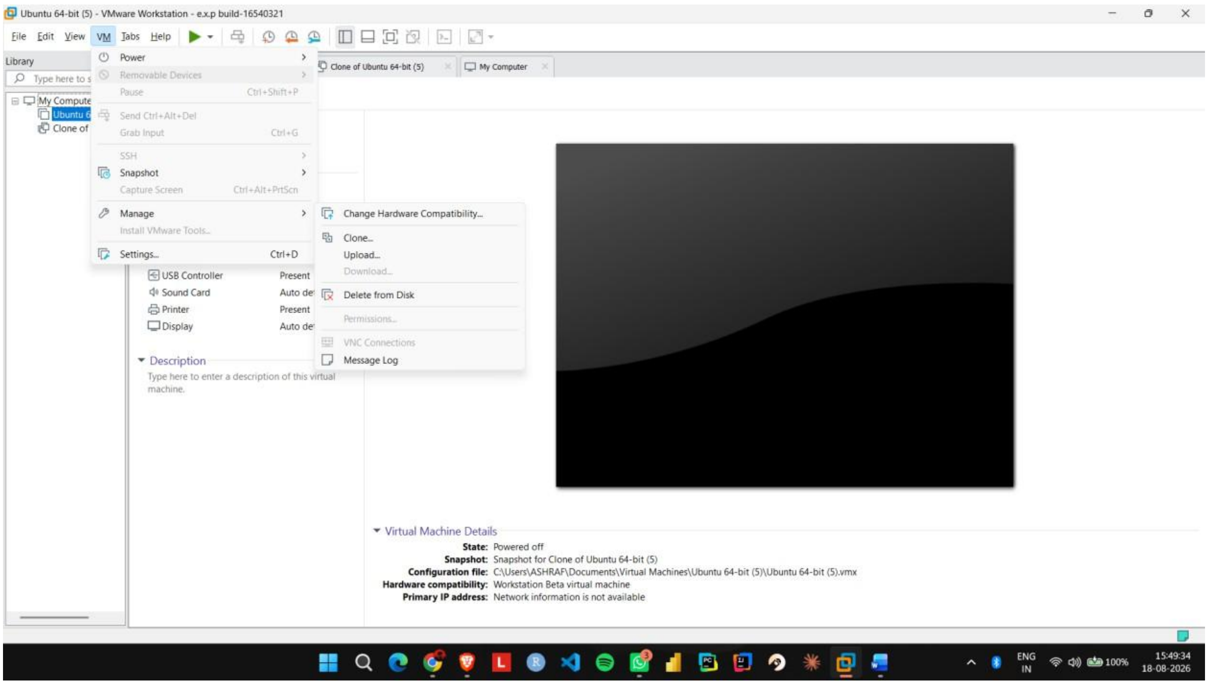
Step 6: Create a test file in the cloned VM and verify that it does not appear in the original Ubuntu VM.

IMPLEMENTATION:

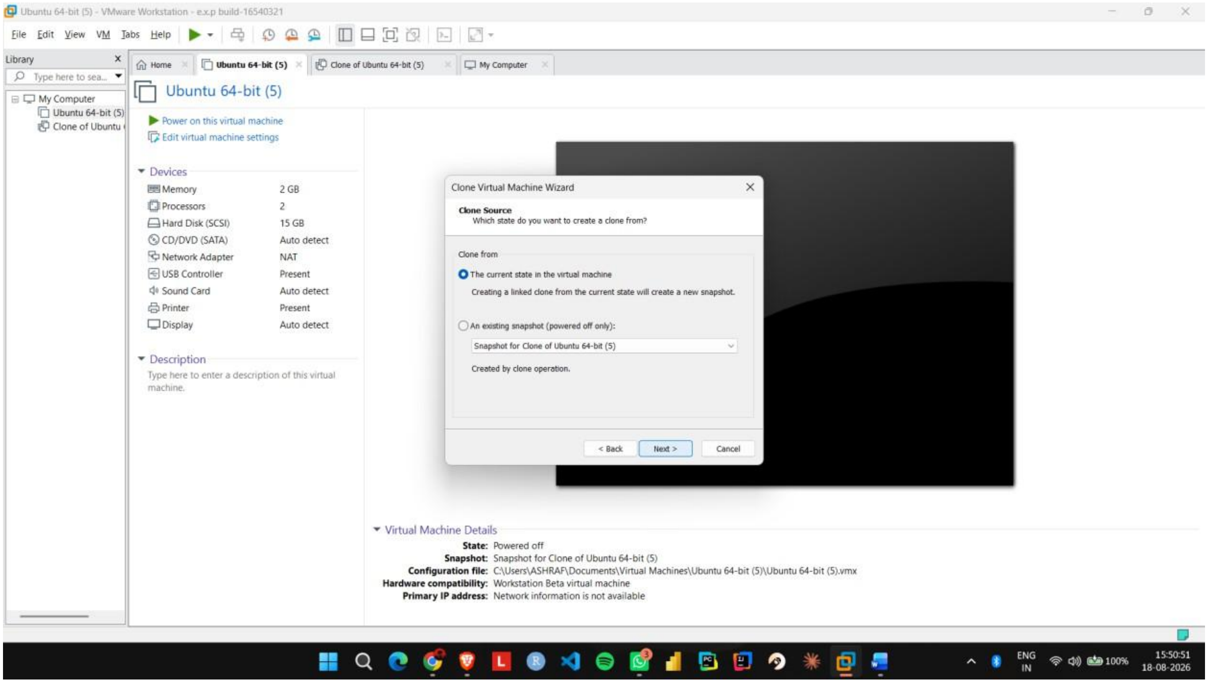
Step 1: Shut down the existing Ubuntu VM completely.



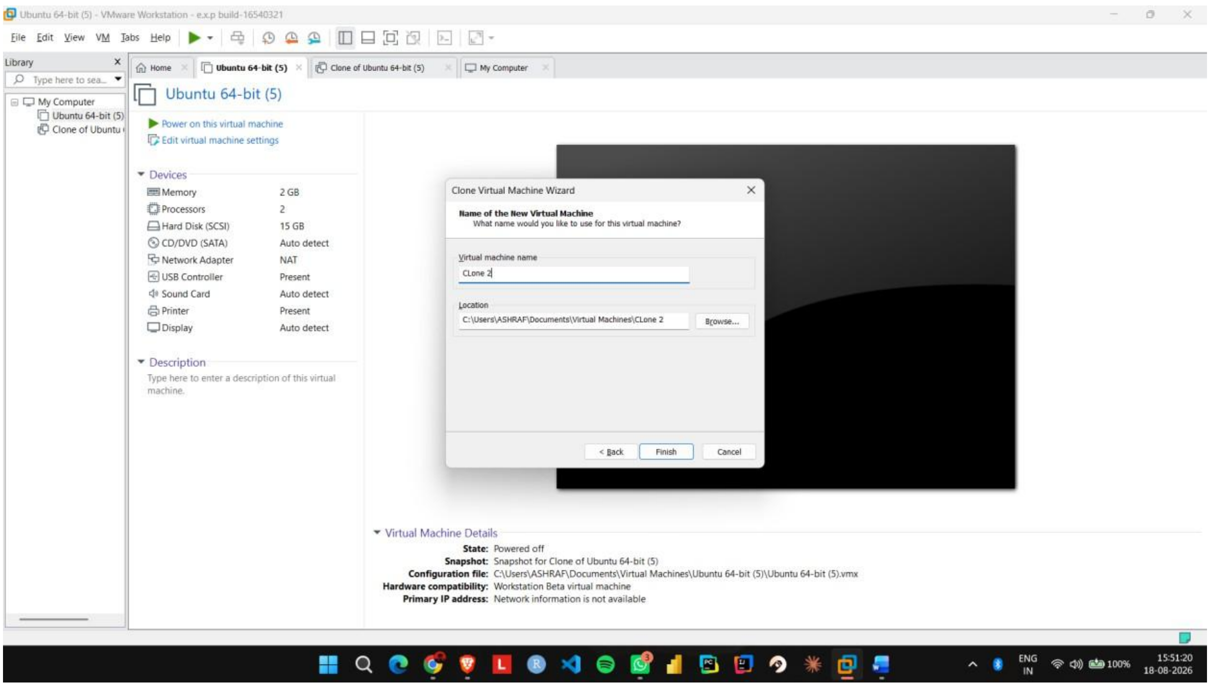
Step 2: Open VMware Workstation and select VM → Manage → Clone.



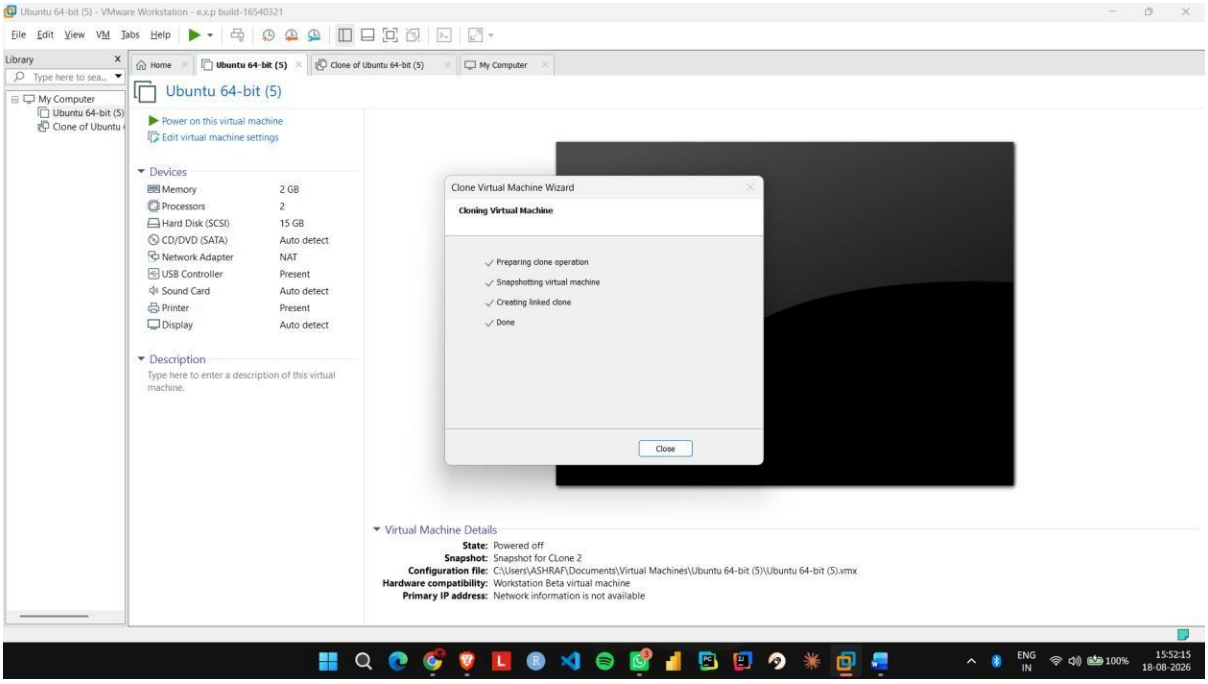
Step 3: Select the current state of the virtual machine as the cloning source.



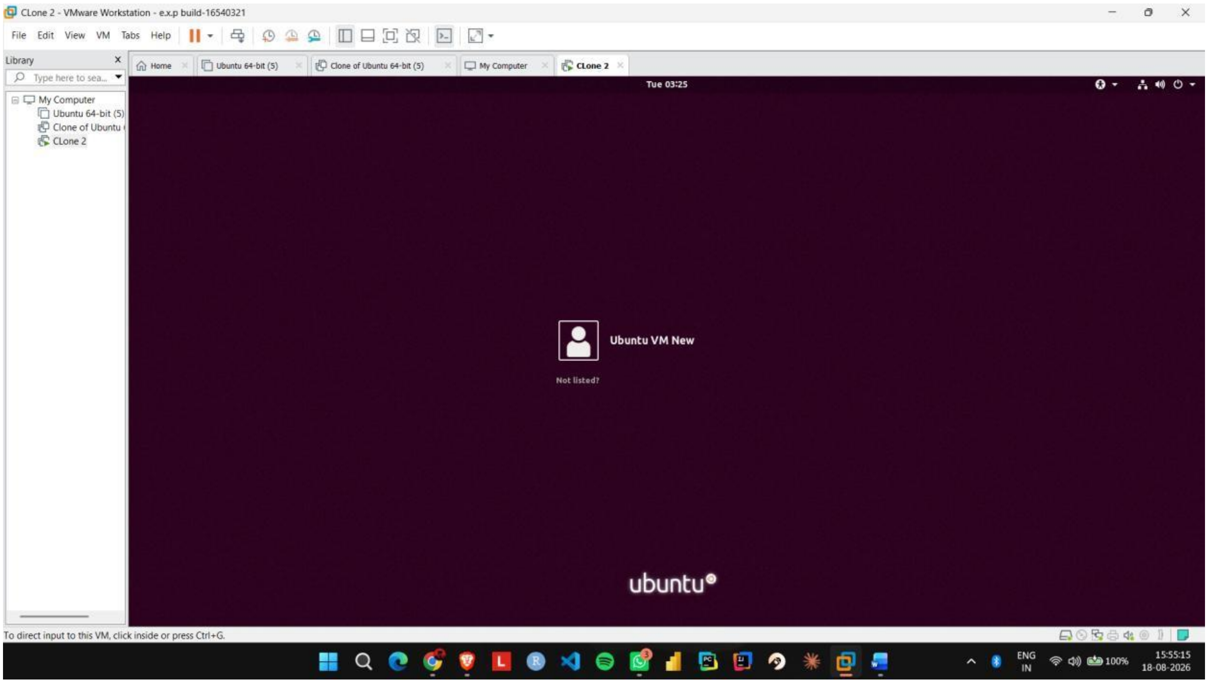
Step 4: Select Create a Full Clone and give the cloned VM a name such as Ubuntu_Clone.



Step 5: Complete the cloning process and start the Ubuntu_Clone virtual machine.



Step 6: Create a test file in the cloned VM and verify that it does not appear in the original Ubuntu VM.



EXP NO 16: Change Hardware compatibility of a VM (Either by clone/create new one) which is already created and configured, using virtual box.

Aim:

To change the hardware configuration and compatibility of an existing virtual machine using VMware Workstation.

Procedure:

Step 1: Shut down the existing Ubuntu VM and open Edit virtual machine settings.

Step 2: Go to Hardware and change the memory from 2 GB to 4 GB.

Step 3: Change the processor configuration from 1 CPU to 2 CPUs.

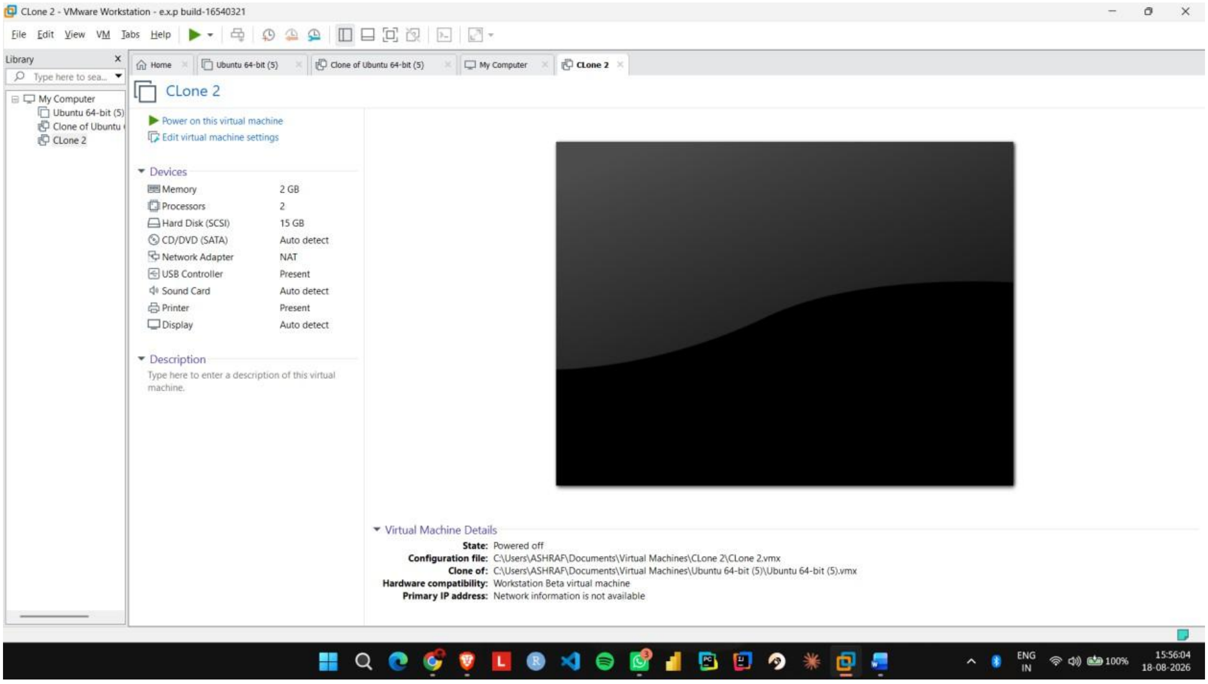
Step 4: Open Options → Advanced and check the virtual machine's hardware compatibility settings.

Step 5: If required, use Manage → Change Hardware Compatibility and select the required VMware hardware version.

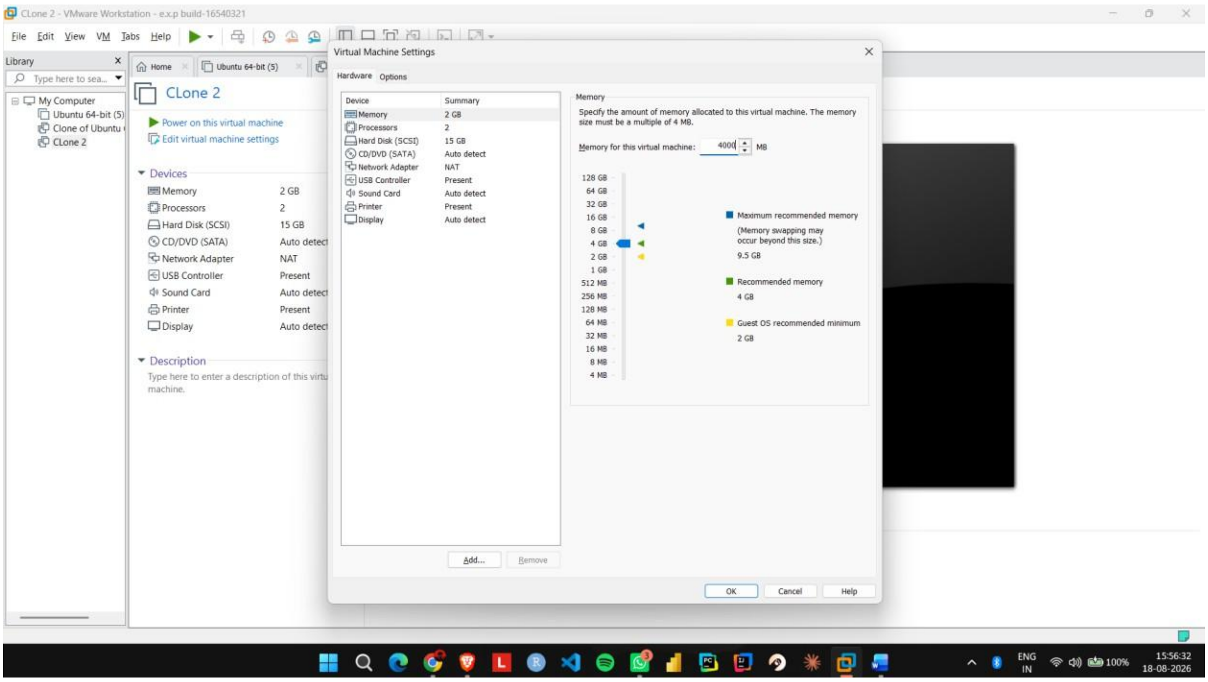
Step 6: Apply the changes, start Ubuntu, and verify that the modified hardware configuration works correctly.

IMPLEMENTATION:

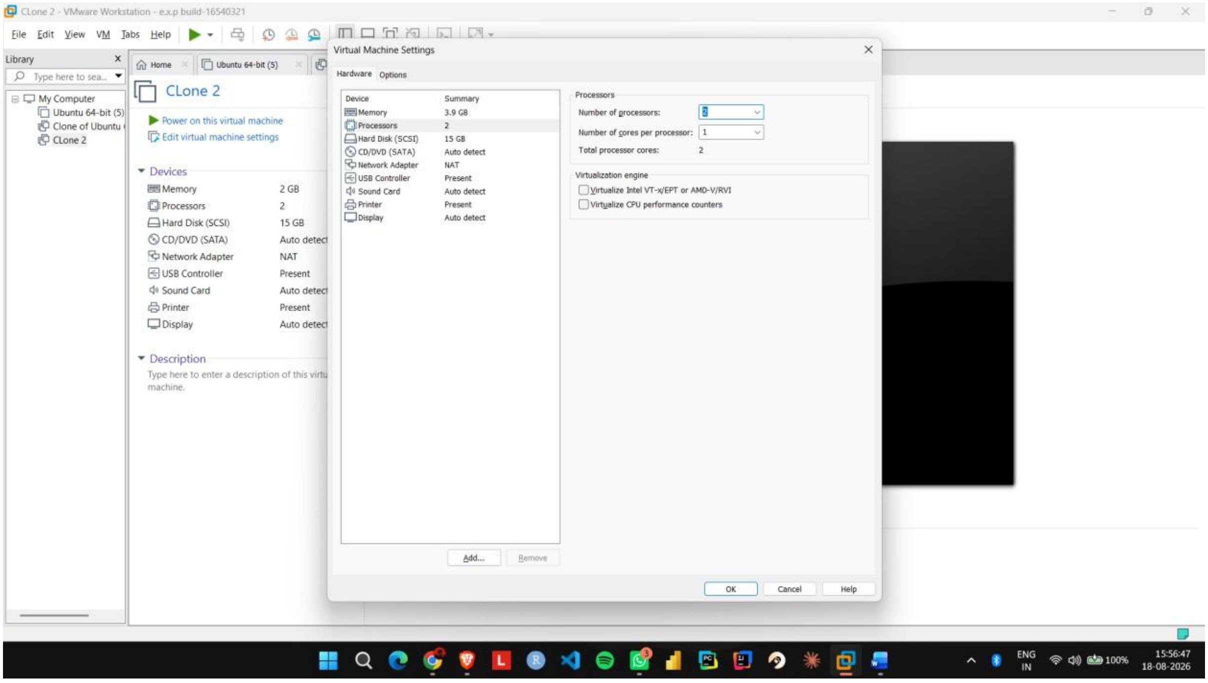
Step 1: Shut down the existing Ubuntu VM and open Edit virtual machine settings.



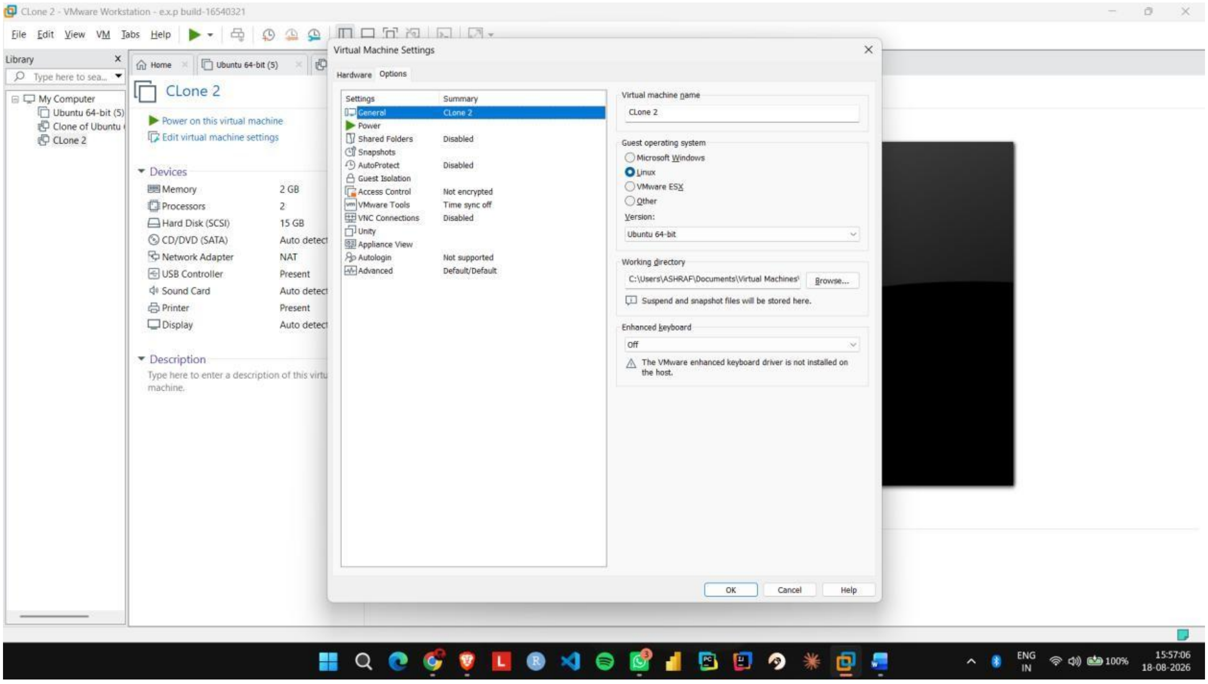
Step 2: Go to Hardware and change the memory from 2 GB to 4 GB.



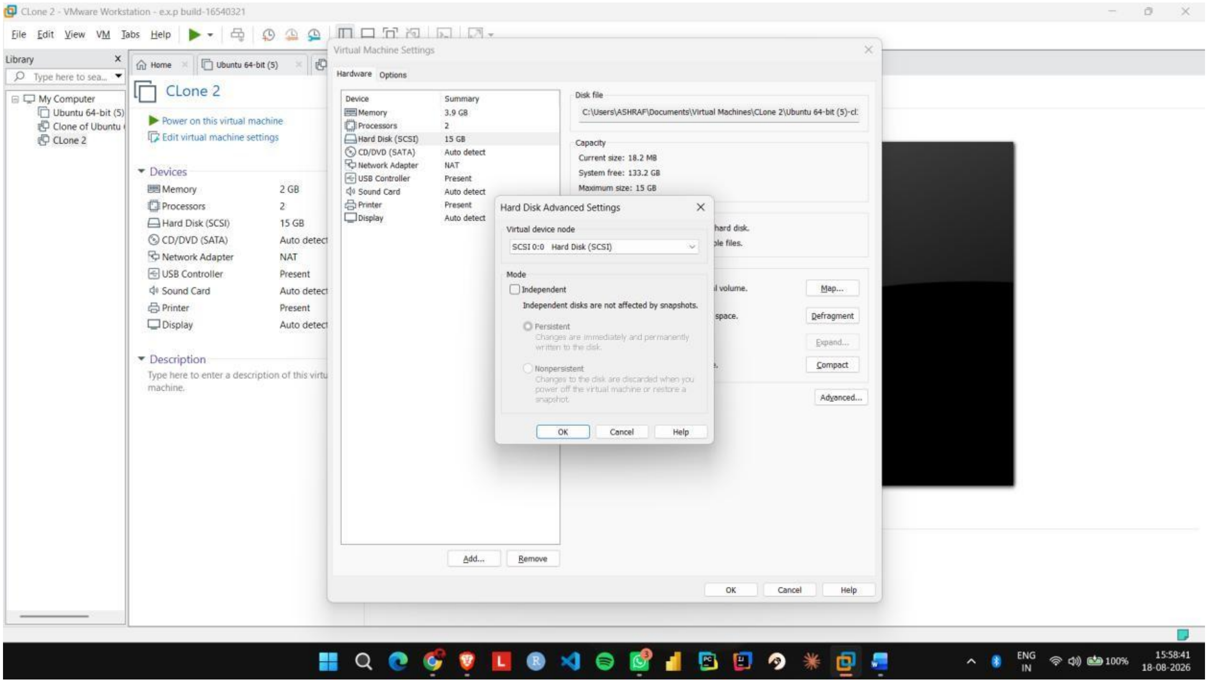
Step 3: Change the processor configuration from 1 CPU to 2 CPUs.



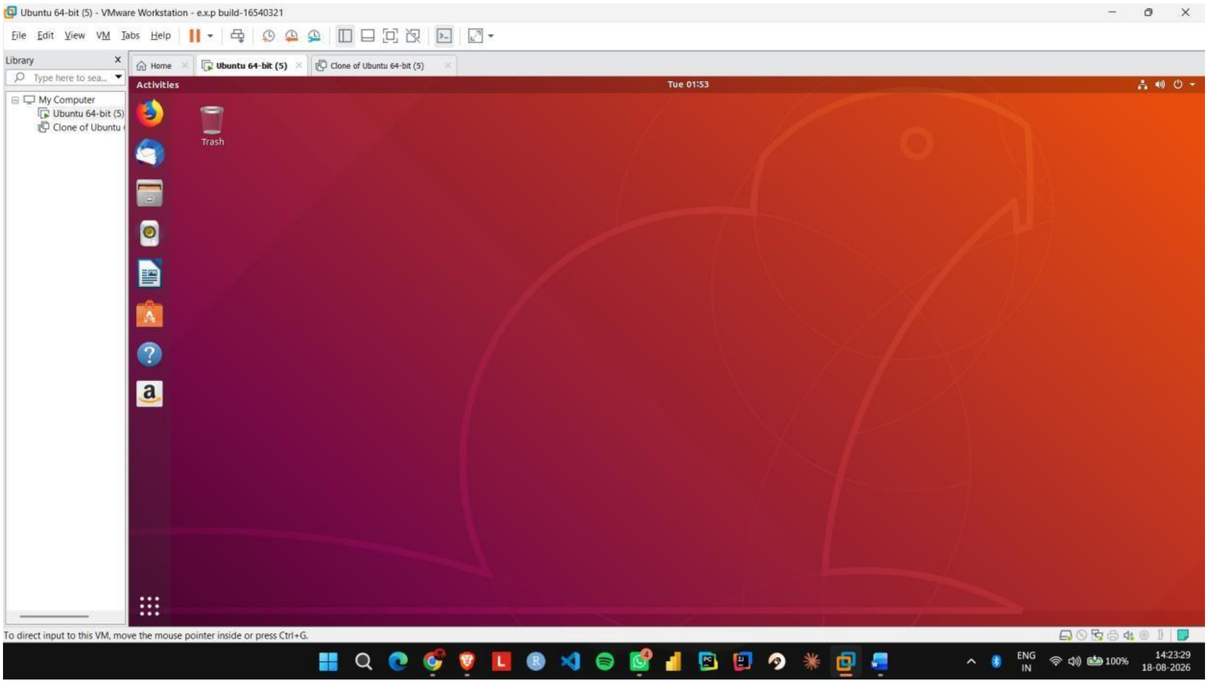
Step 4: Open Options → Advanced and check the virtual machine's hardware compatibility settings.



Step 5: If required, use Manage → Change Hardware Compatibility and select the required VMware hardware version.



Step 6: Apply the changes, start Ubuntu, and verify that the modified hardware configuration works.



EXP NO 17: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (EitherWindows/Linux), using VMware workstation.

Aim:

To demonstrate Type-2 virtualization by installing VMware Workstation and creating a configured Ubuntu virtual machine.

Procedure:

Step 1: Install and open VMware Workstation on the host operating system.

Step 2: Select Create a New Virtual Machine and choose the Typical configuration option.

Step 3: Select the Ubuntu ISO image and provide a suitable name for the virtual machine.

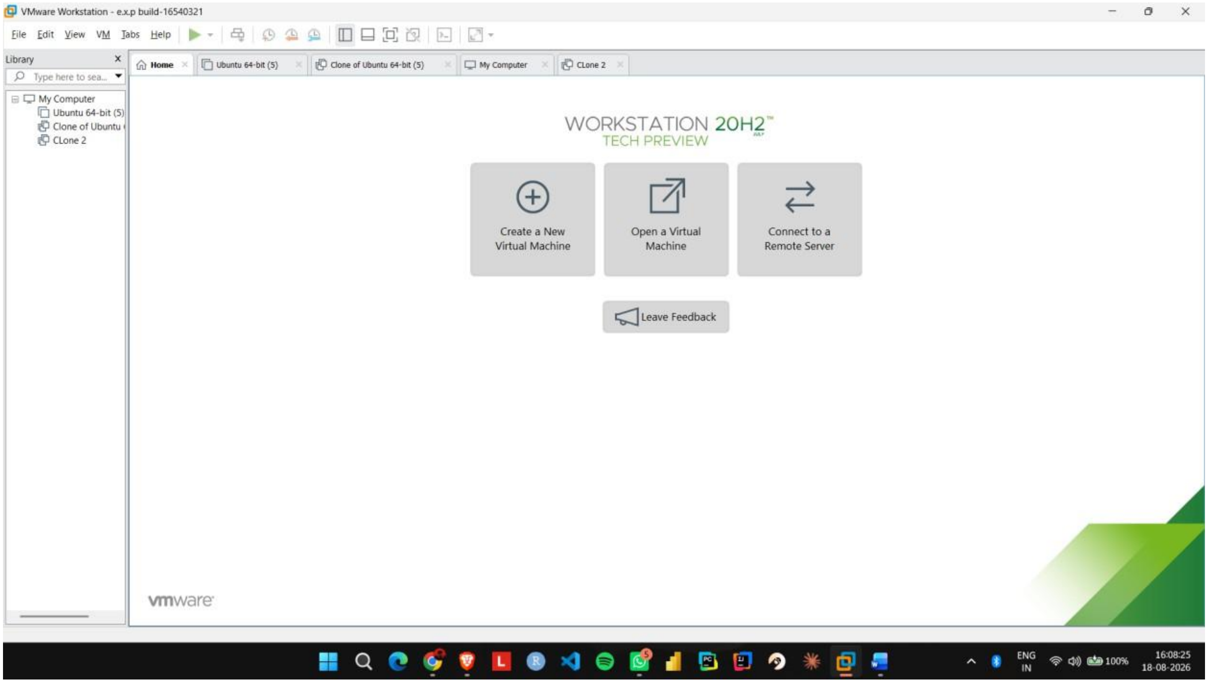
Step 4: Open Customize Hardware and configure 1 CPU, 2 GB RAM, and 15 GB disk storage.

Step 5: Select NAT for the network adapter and complete the virtual machine configuration.

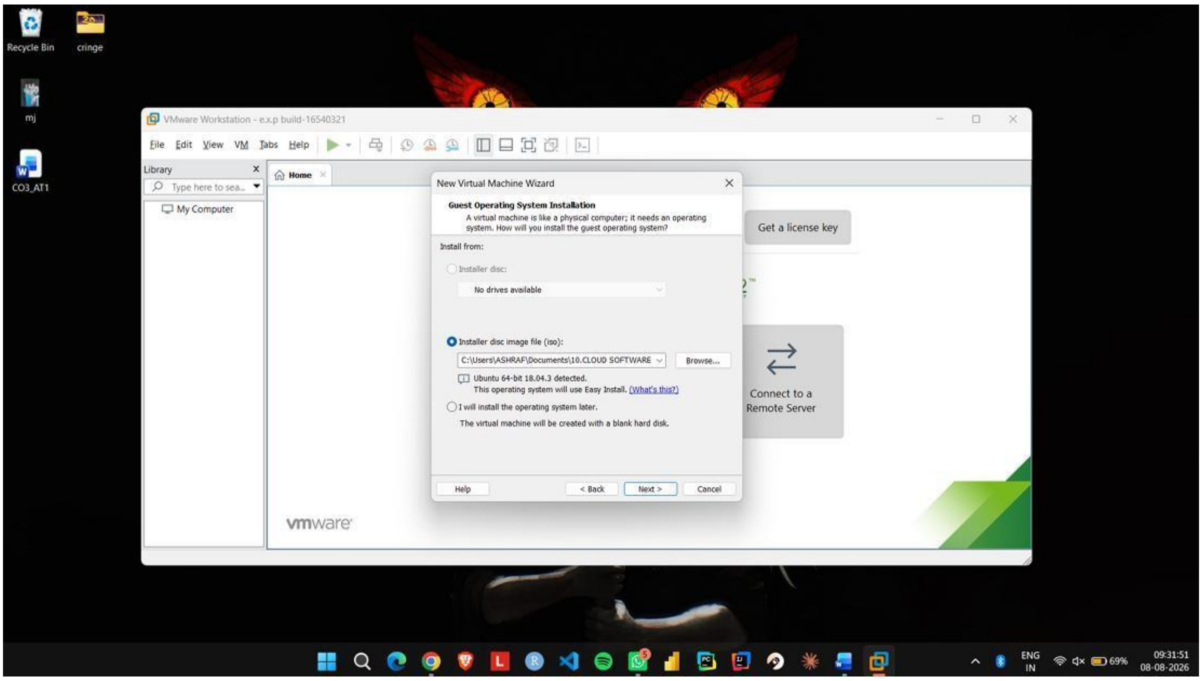
Step 6: Start the VM, install Ubuntu, and verify that the guest operating system runs successfully inside VMware Workstation.

IMPLEMENTATION:

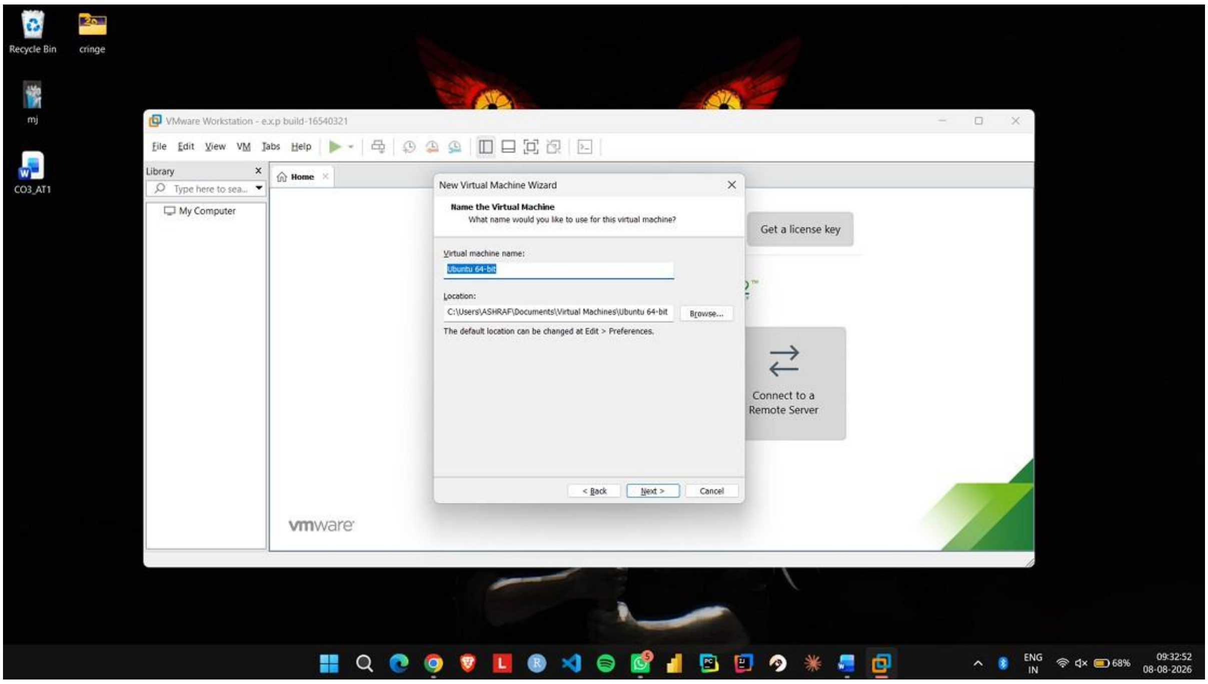
Step 1: Install and open VMware Workstation on the host operating system.



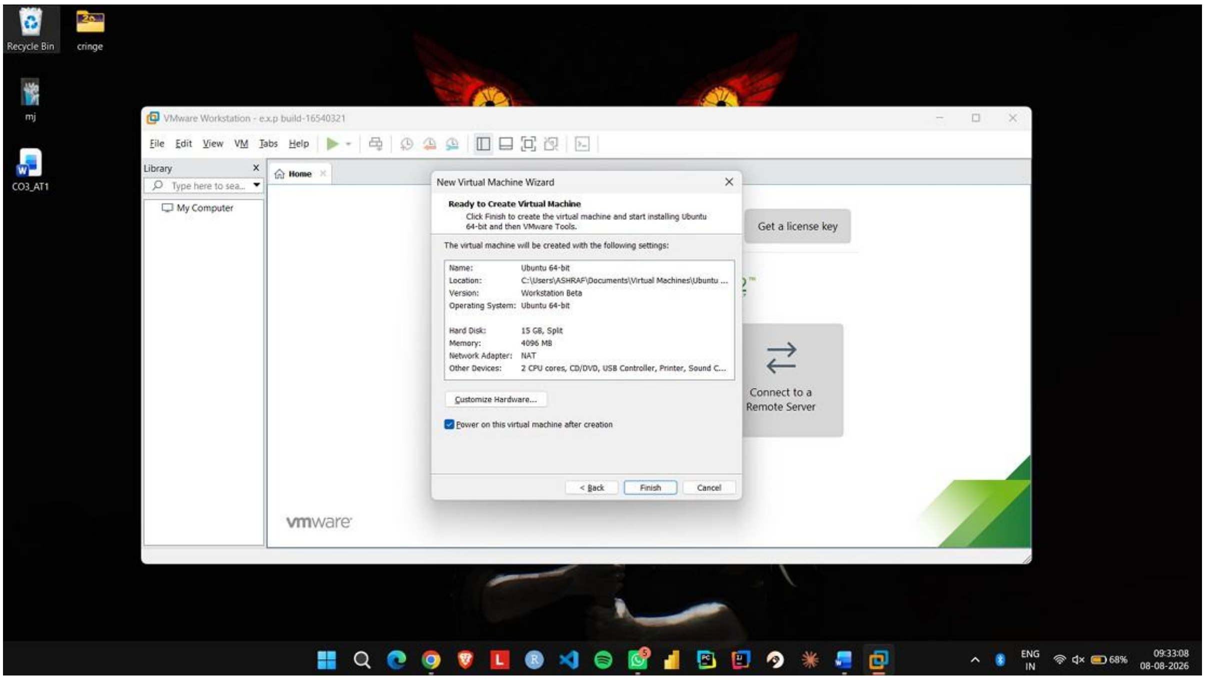
Step 2: Select Create a New Virtual Machine and choose the Typical configuration option.



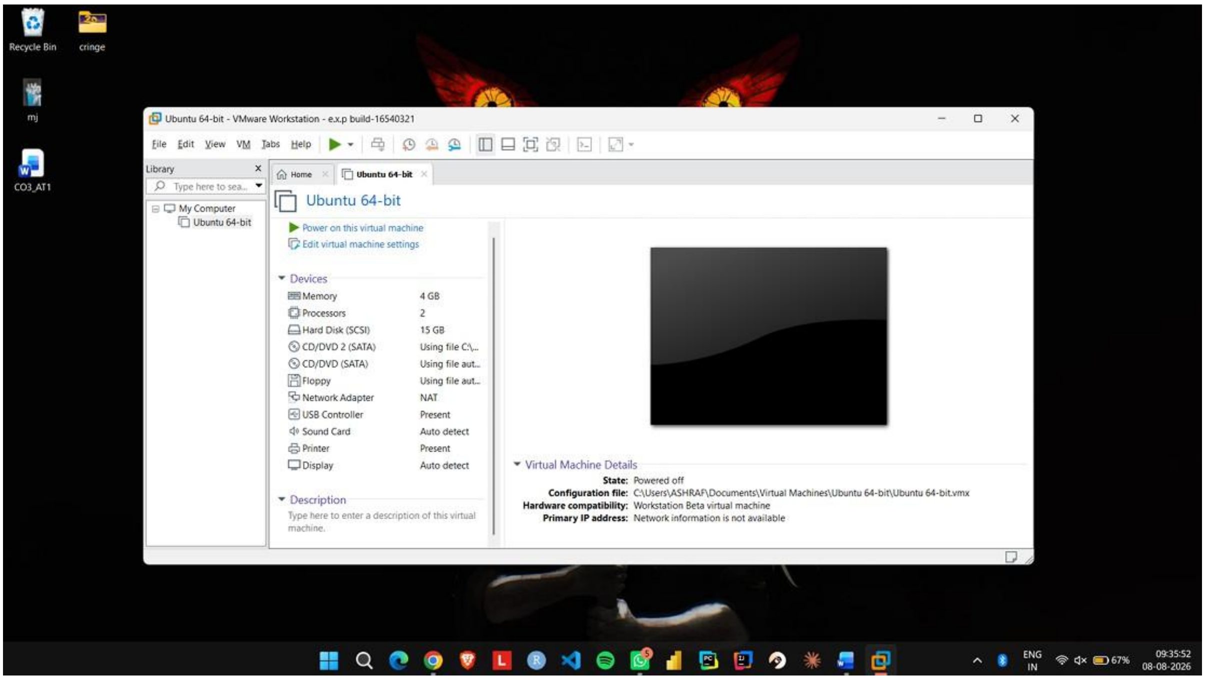
Step 3: Select the Ubuntu ISO image and provide a suitable name for the virtual machine.



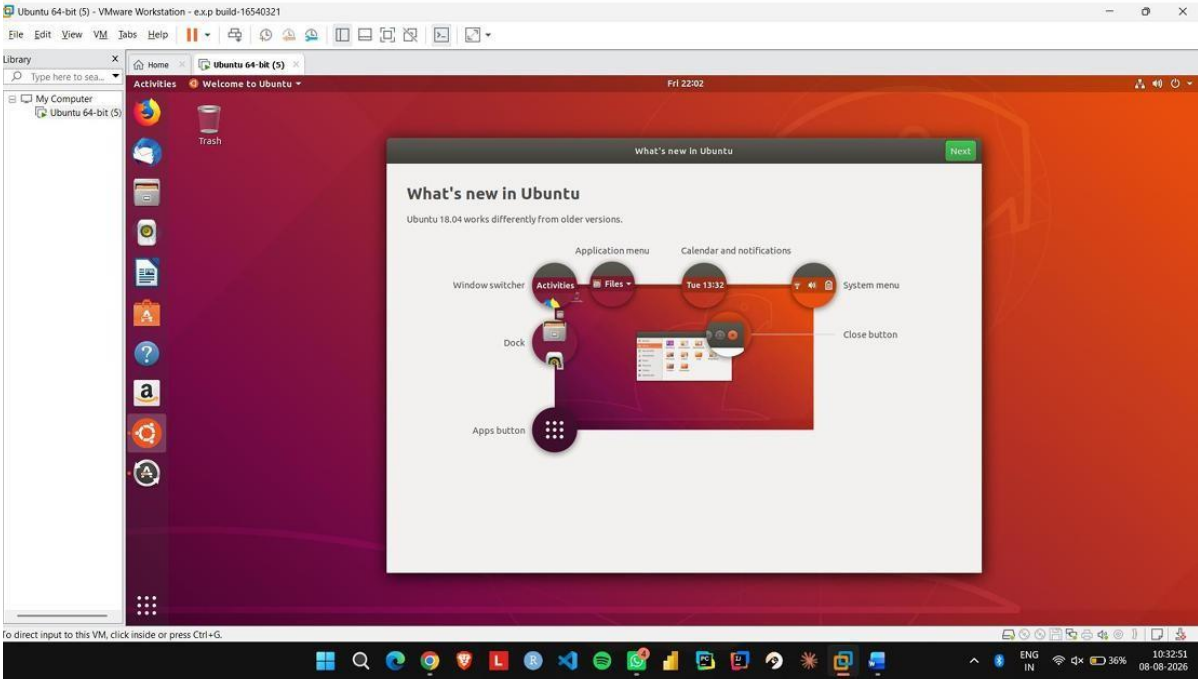
Step 4: Open Customize Hardware and configure 1 CPU, 2 GB RAM, and 15 GB disk storage.



Step 5: Select NAT for the network adapter and complete the virtual machine configuration.



Step 6: Start the VM, install Ubuntu, and verify that the guest operating system runs successfully inside VMware Workstation.



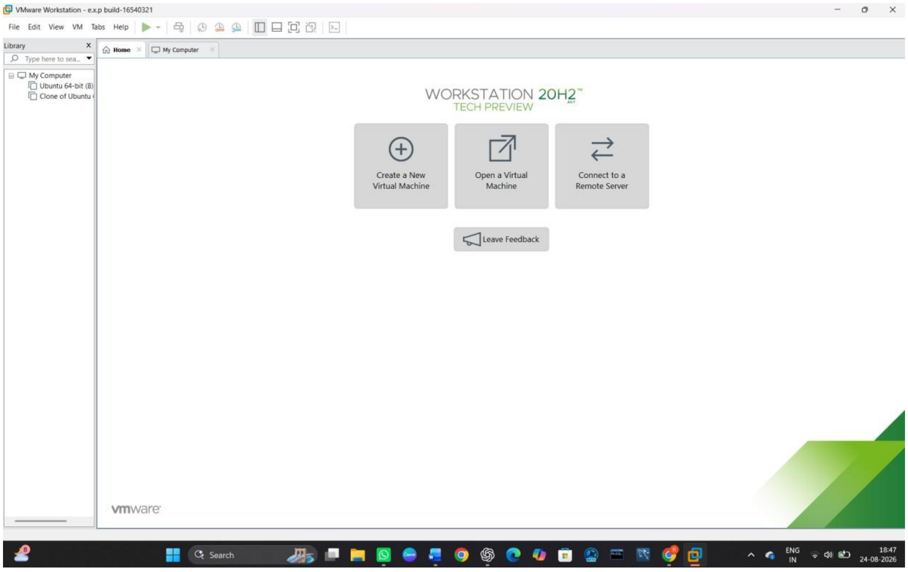
EXP NO18: Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of your present windows environment.

AIM:

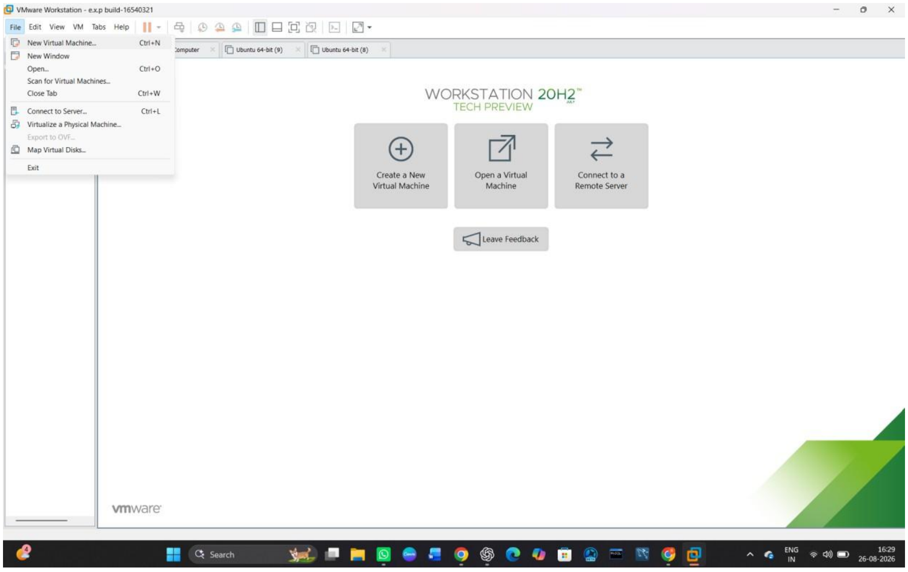
To install VMware Workstation and install a different flavour of Linux/Windows Operating System as a virtual machine on the existing Windows environment.

PROCEDURE:

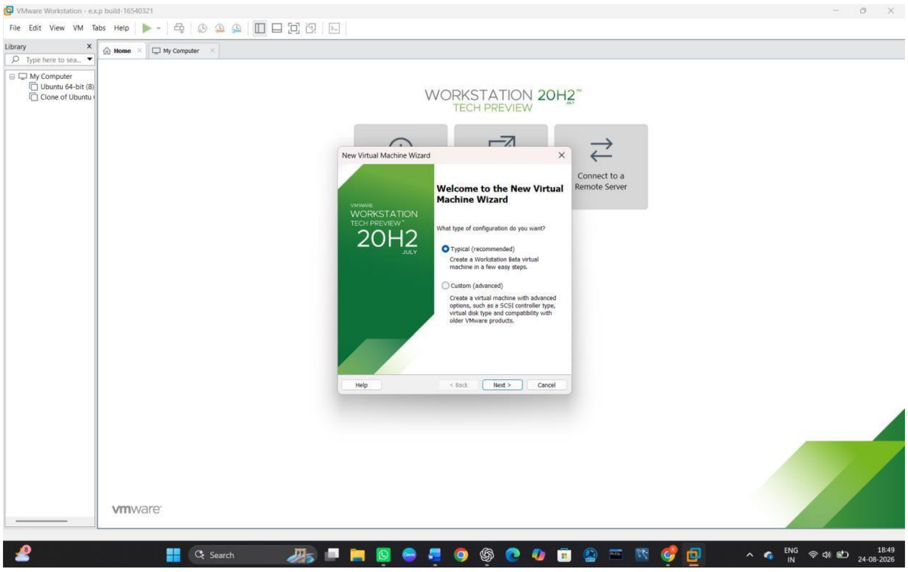
STEP 1: Go to VMware Workstation.



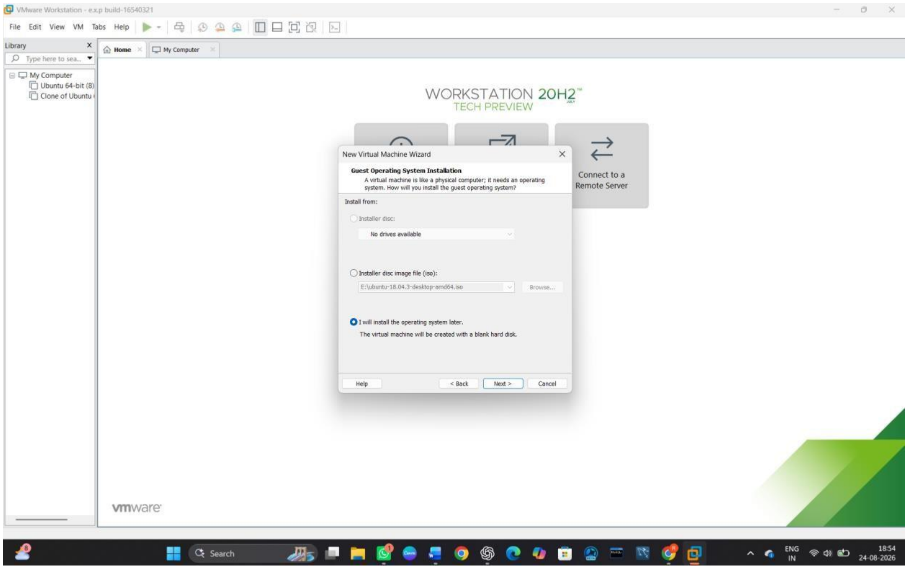
STEP 2: Click on Create a New Virtual Machine.

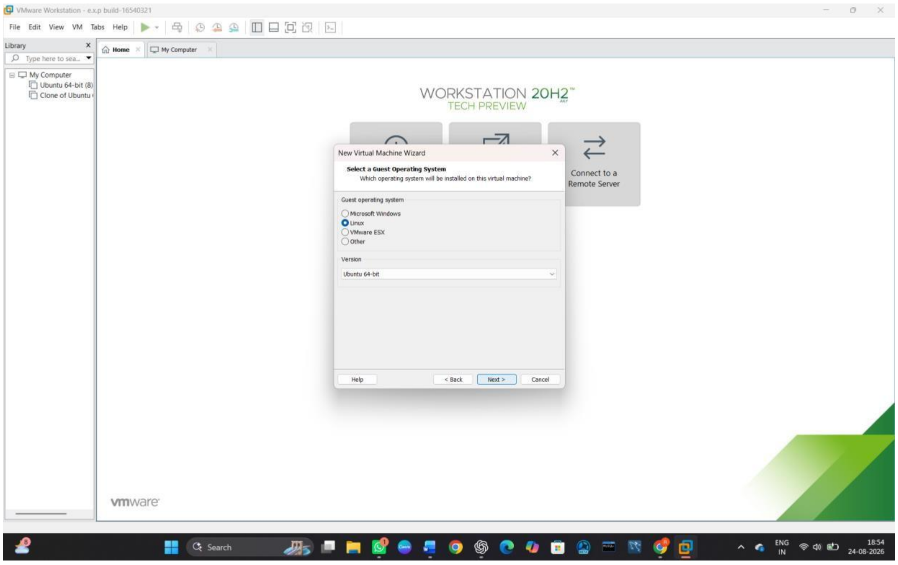


STEP 3: Select Typical (Recommended) and click Next.

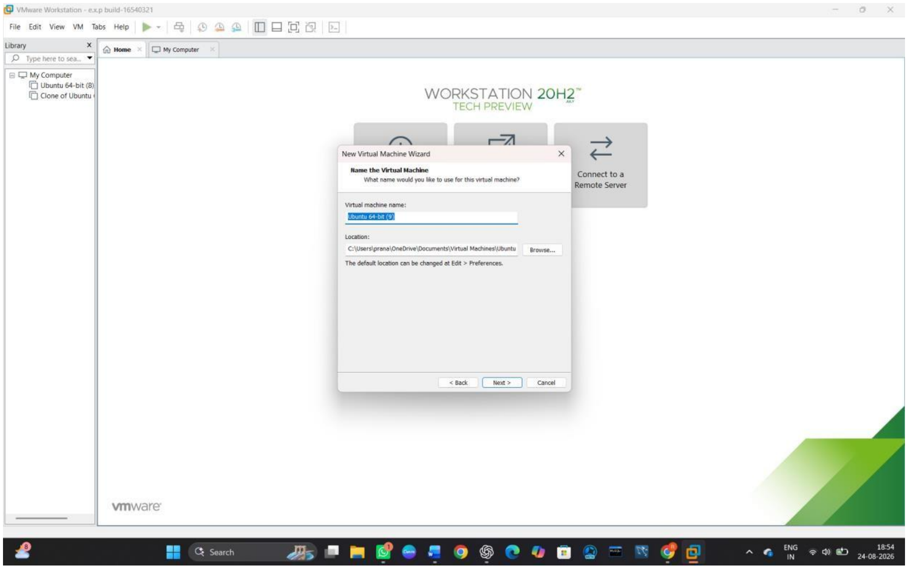


STEP 4: Select the ISO image file of the Linux/Windows operating system.

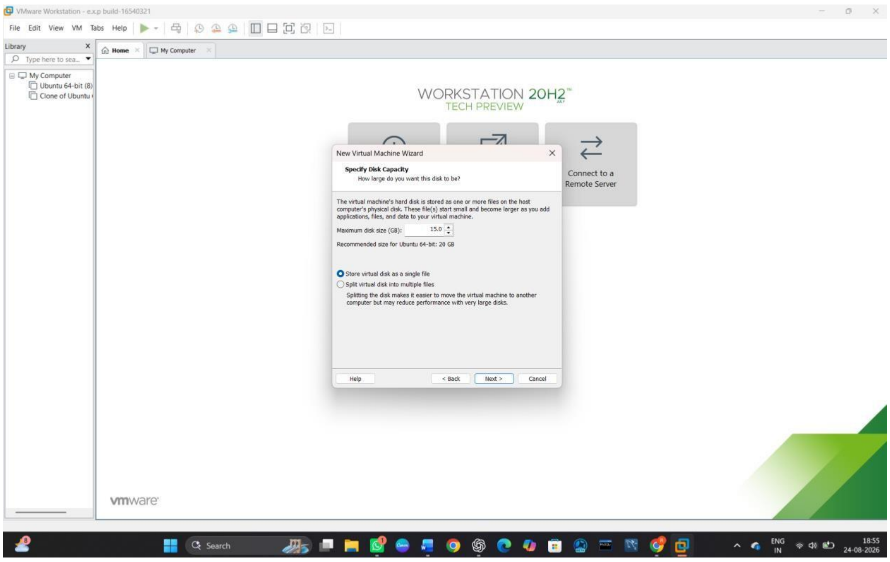




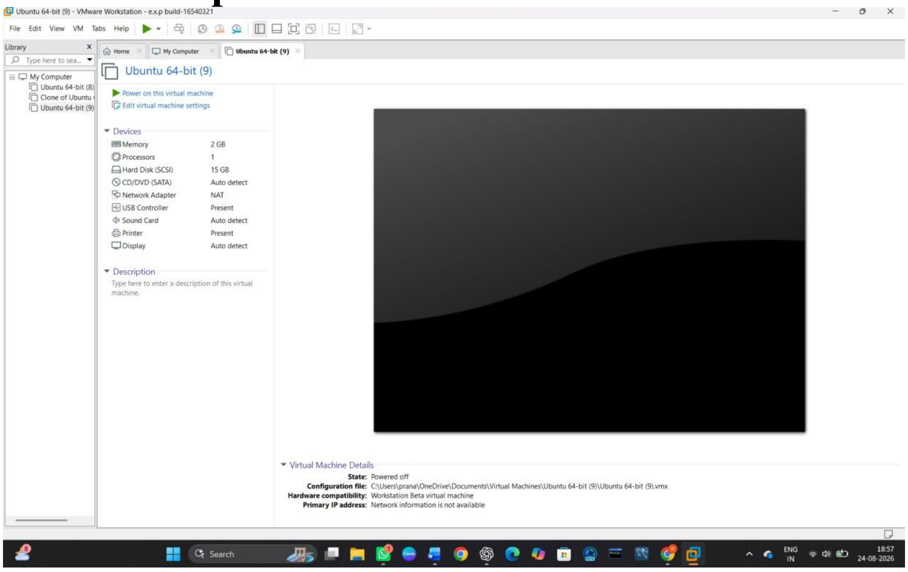
STEP 5: Enter the **Virtual Machine Name** and select the storage location.



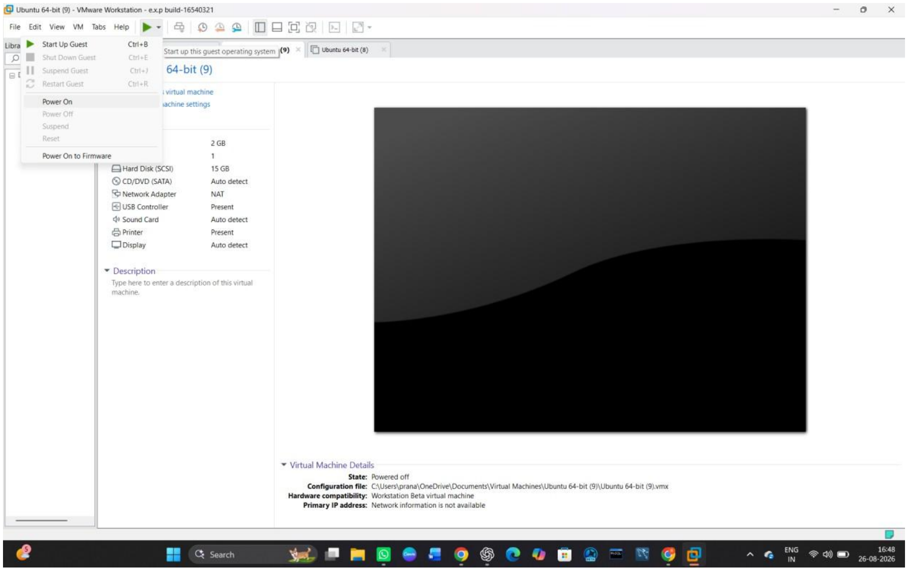
STEP 6: Specify the required **disk size** and click **Next**.

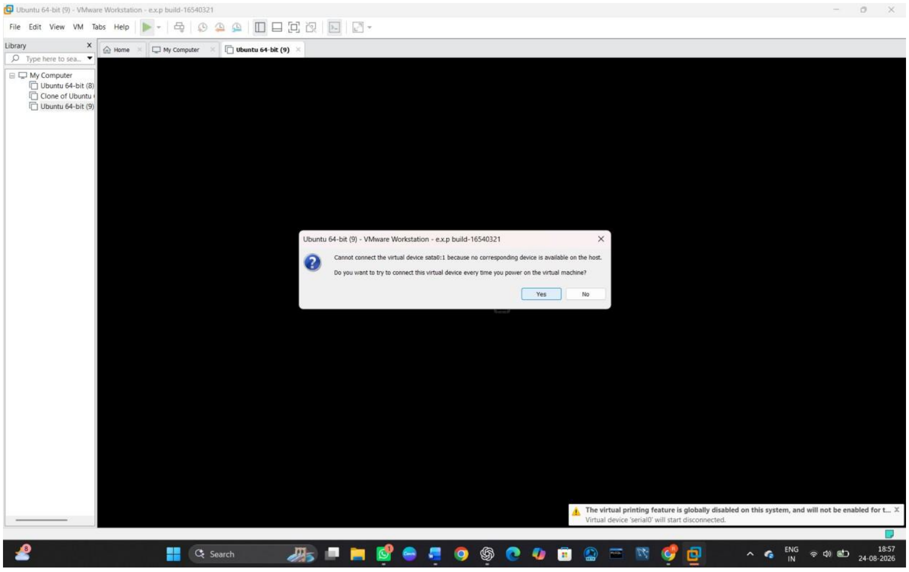


STEP 7: Click on Customize Hardware and configure the required RAM and processor.

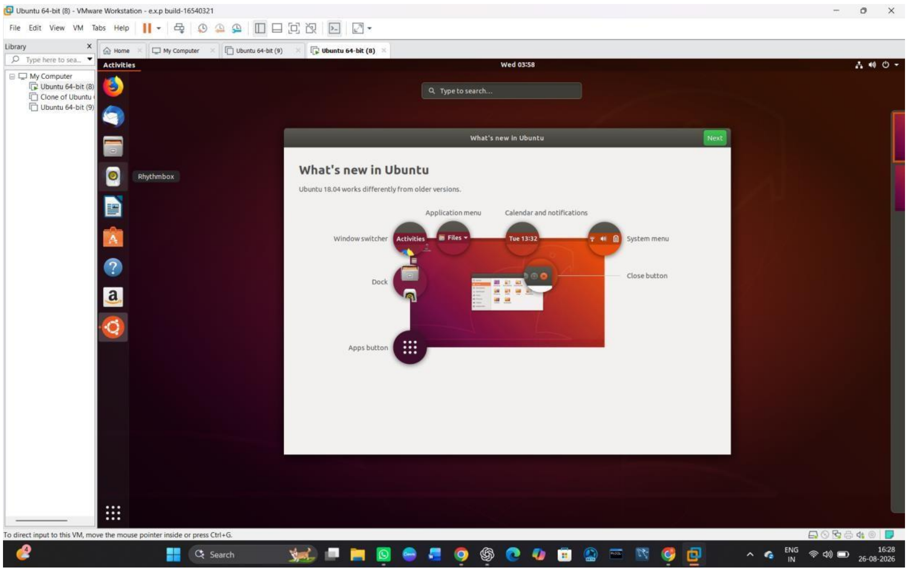


STEP 8: Click **Power on this virtual machine**.





STEP 9: Install the selected Linux/Windows operating system by following the installation instructions.



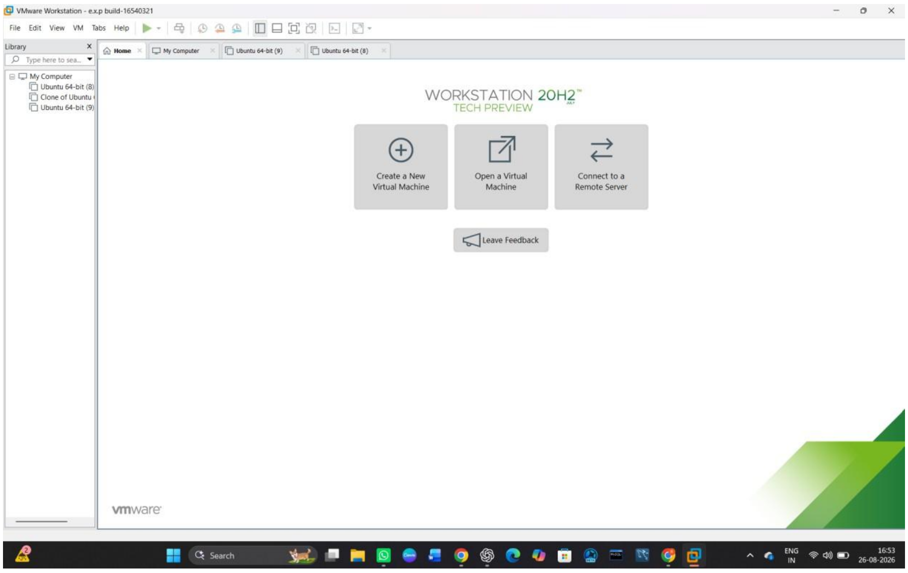
EXP NO19:Install a C compiler in the virtual machine created using virtual box / VM Ware Workstation / Player and execute Simple C Programs.

AIM:

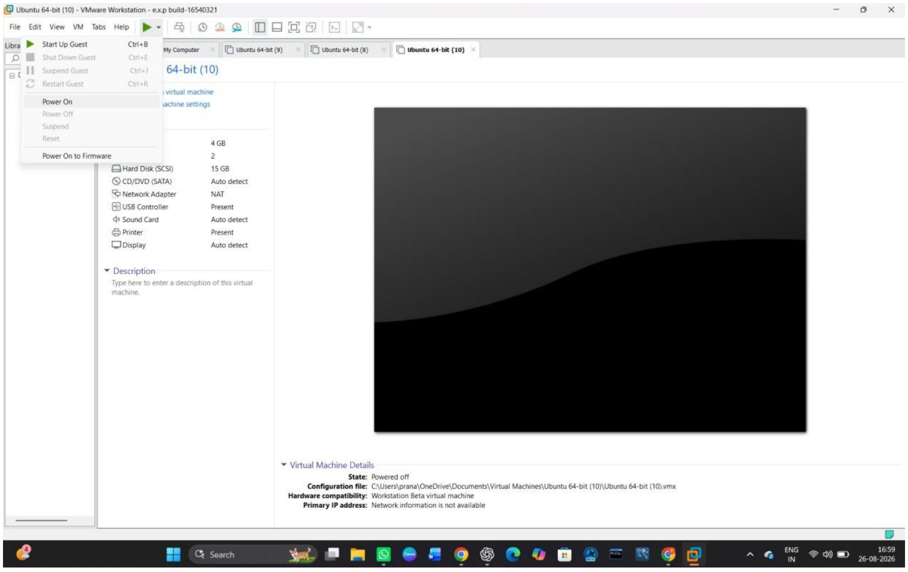
To install a **C compiler (GCC)** in the virtual machine created using **VMware Workstation** and execute simple C programs.

PROCEDURE:

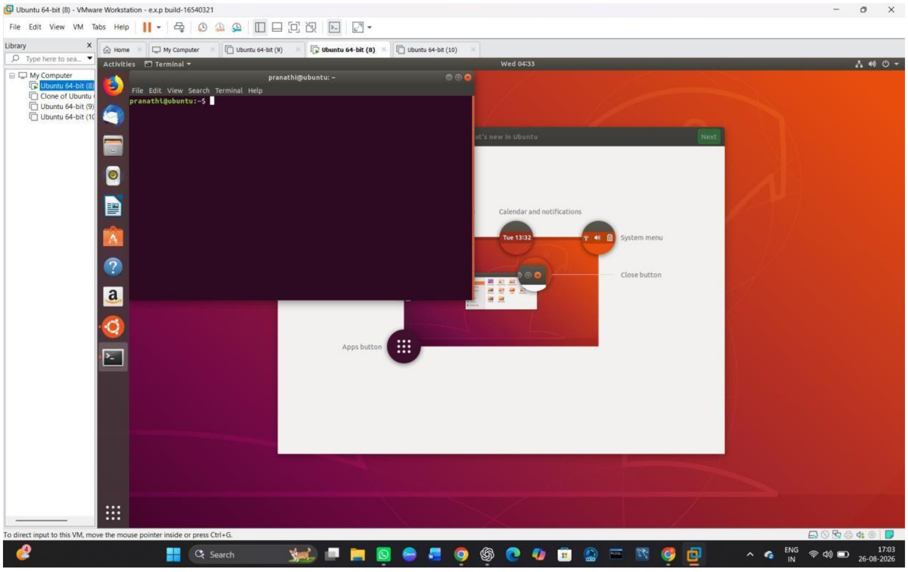
STEP 1: Go to **VMware Workstation**.



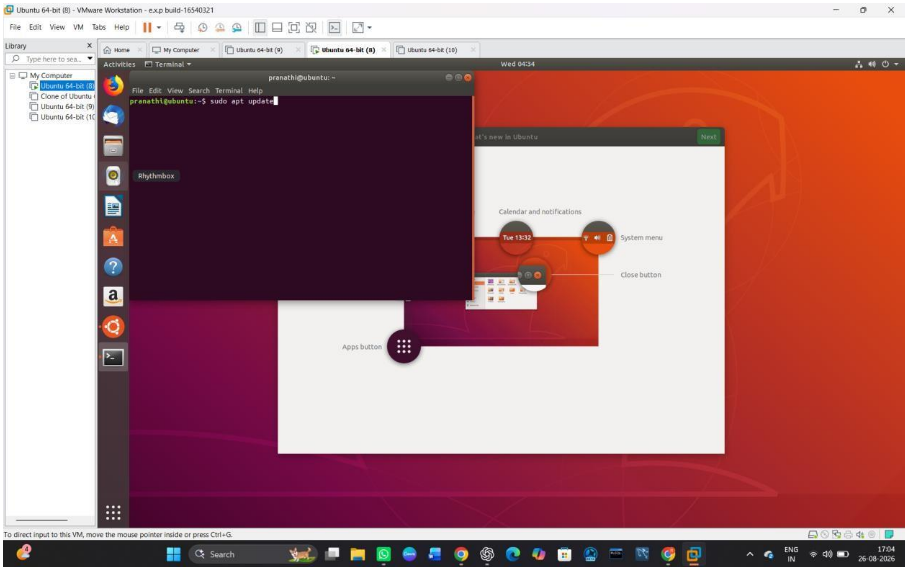
STEP 2: Start the Ubuntu virtual machine.



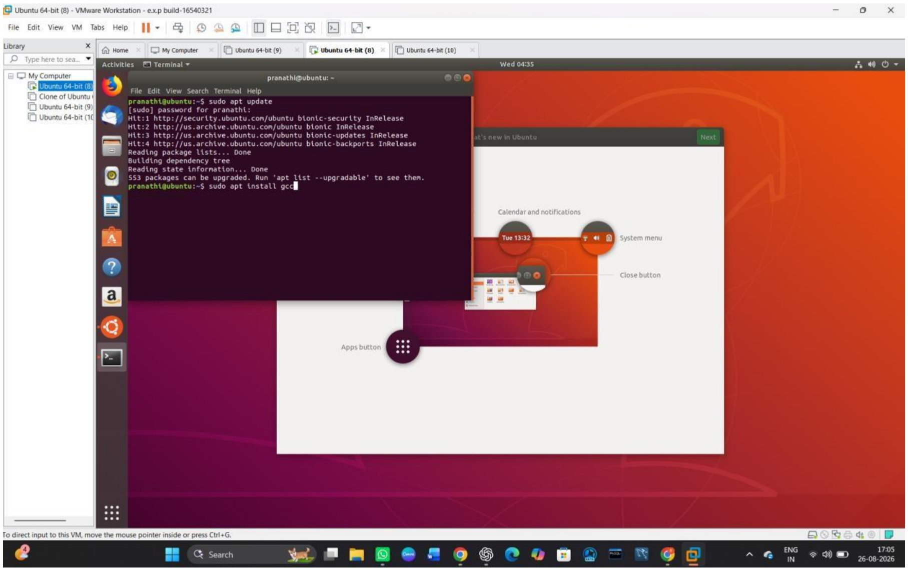
STEP 3: Open the **Terminal** in Ubuntu.



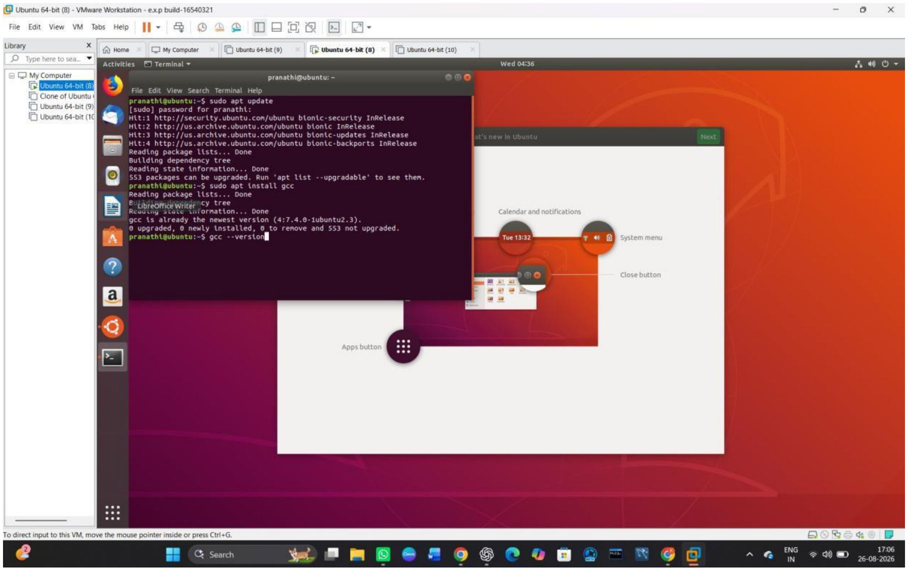
STEP 4: Update the package list using the command:



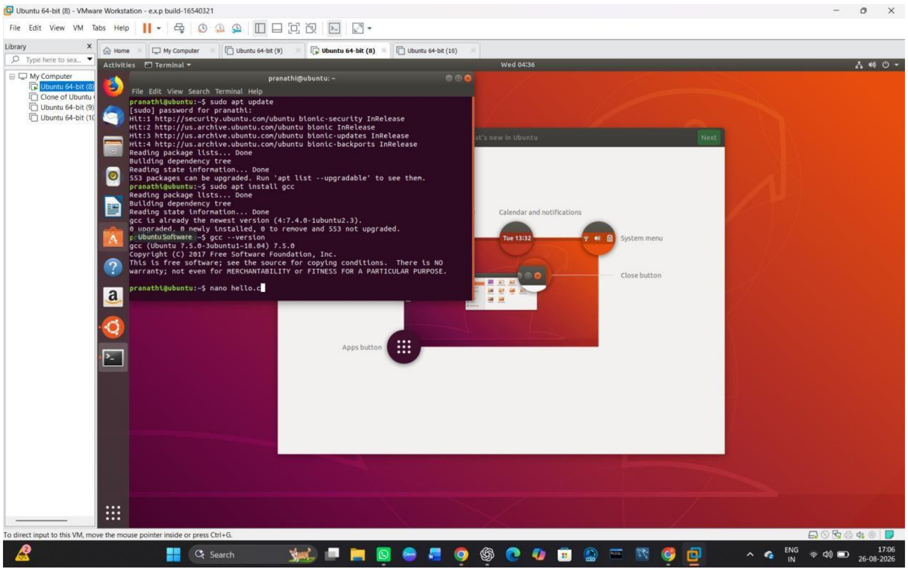
STEP 5: Install the GCC compiler using:



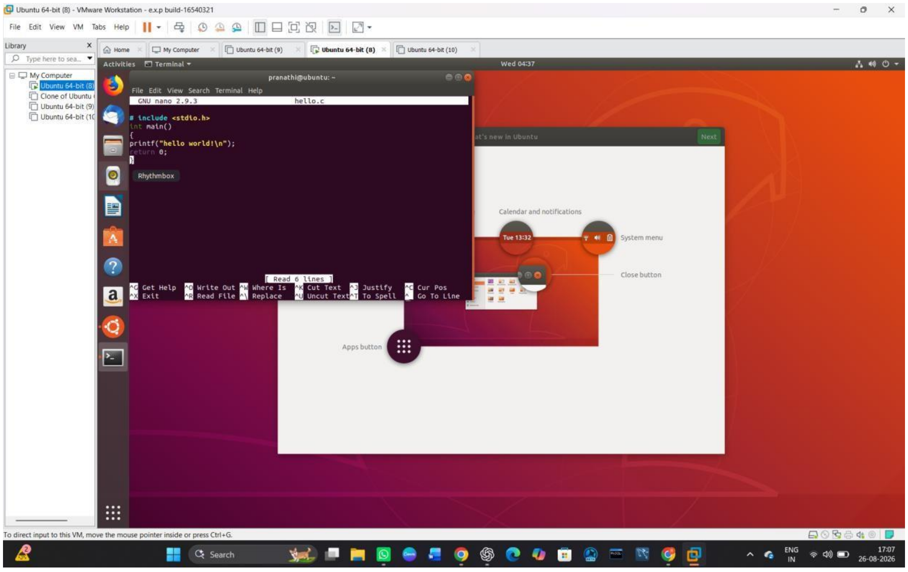
STEP 6: Check whether GCC is installed using:



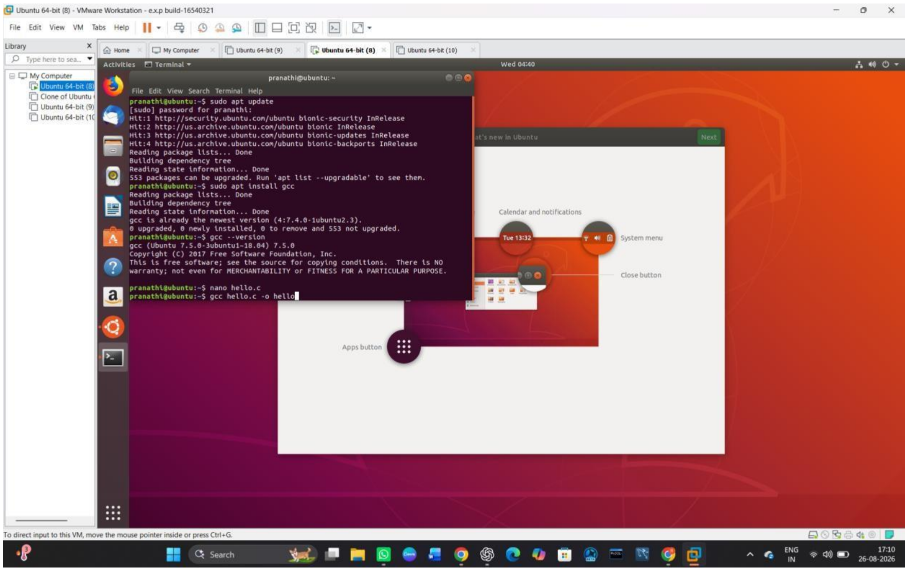
STEP 7: Create a C program using:



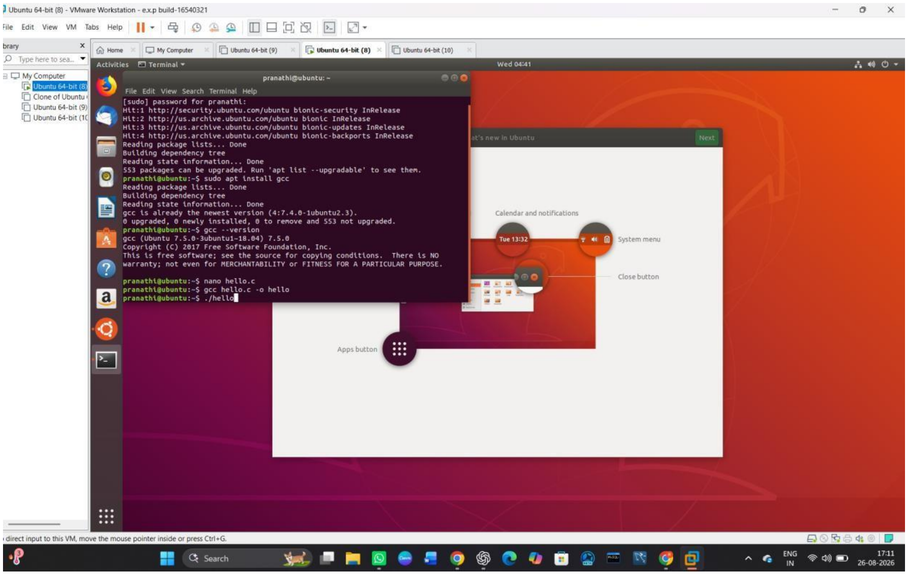
STEP 8: Enter a simple C program and save the file.



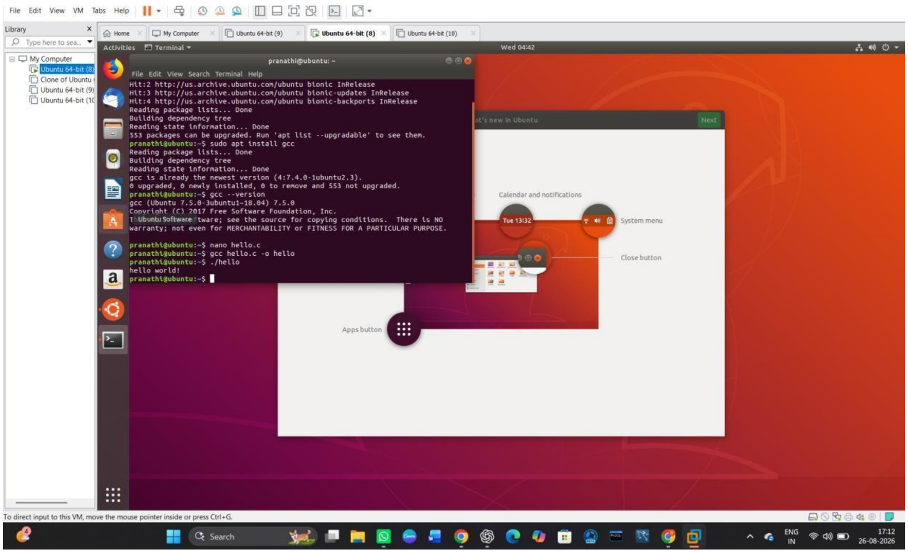
STEP 9: Compile the program using:



STEP 10: Execute the program using:



STEP 11: Succesfully Executed C program



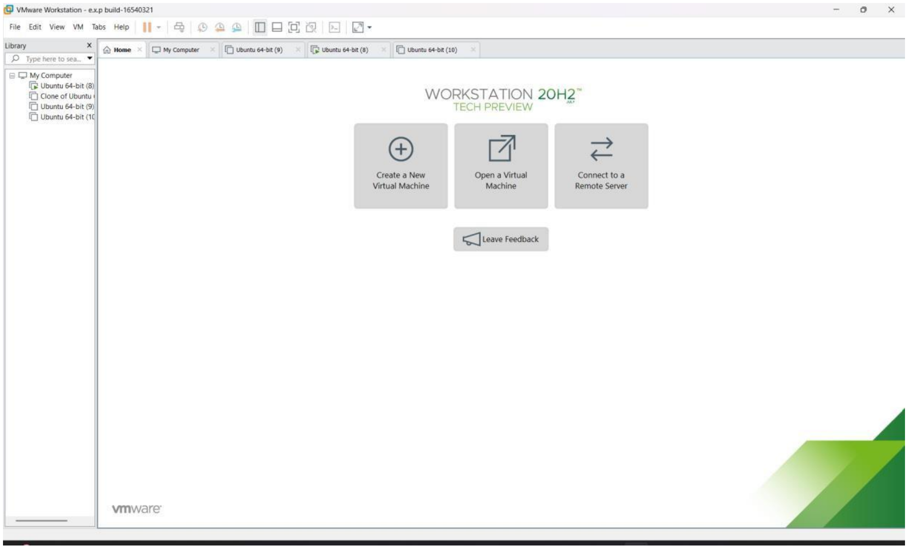
EXP NO 20:Install Virtualbox with different flavours of linux or windows OS on top of windows 10 using custom installation.

AIM:

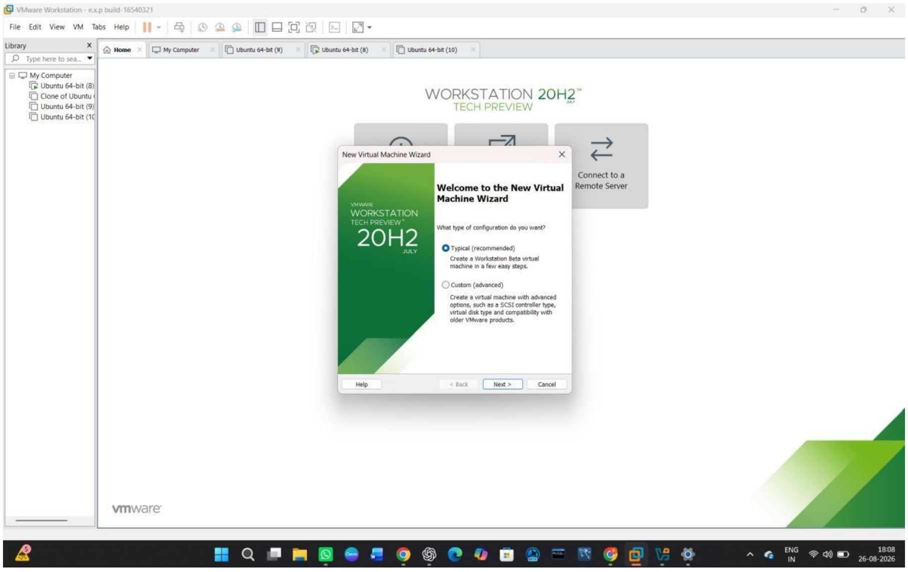
To install Oracle VirtualBox with a different flavour of Linux/Windows Operating System on top of the existing Windows 10 environment using Custom Installation.

PROCEDURE:

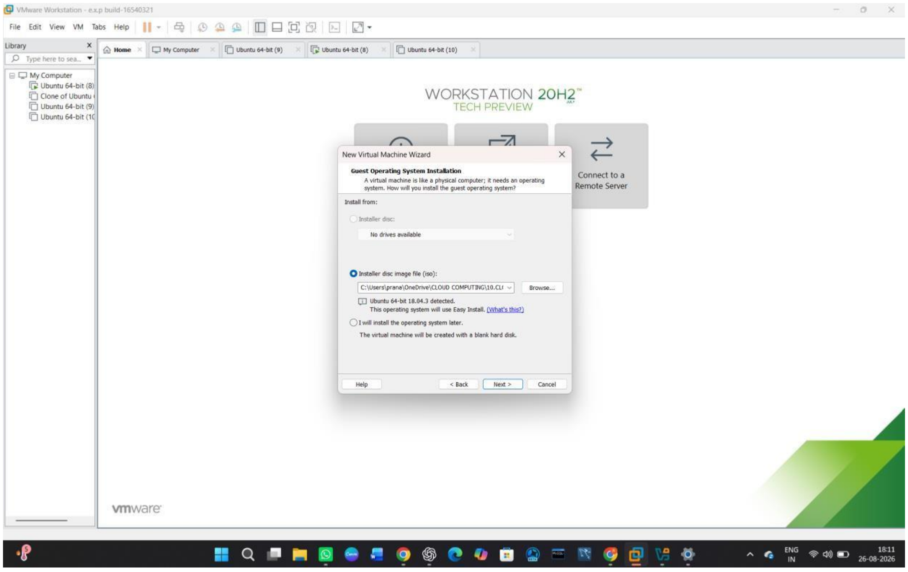
STEP 1: Open VMware Workstation



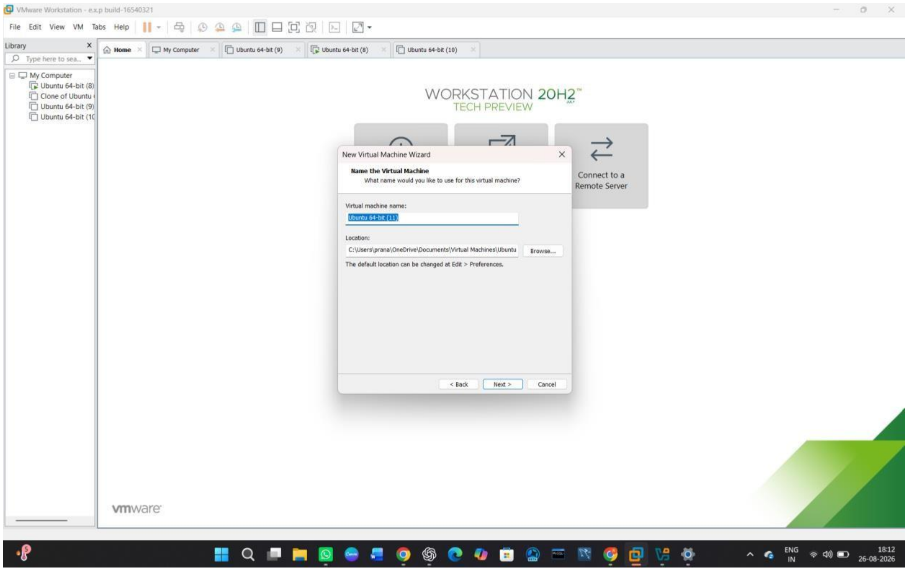
STEP 2: Create a New Virtual Machine



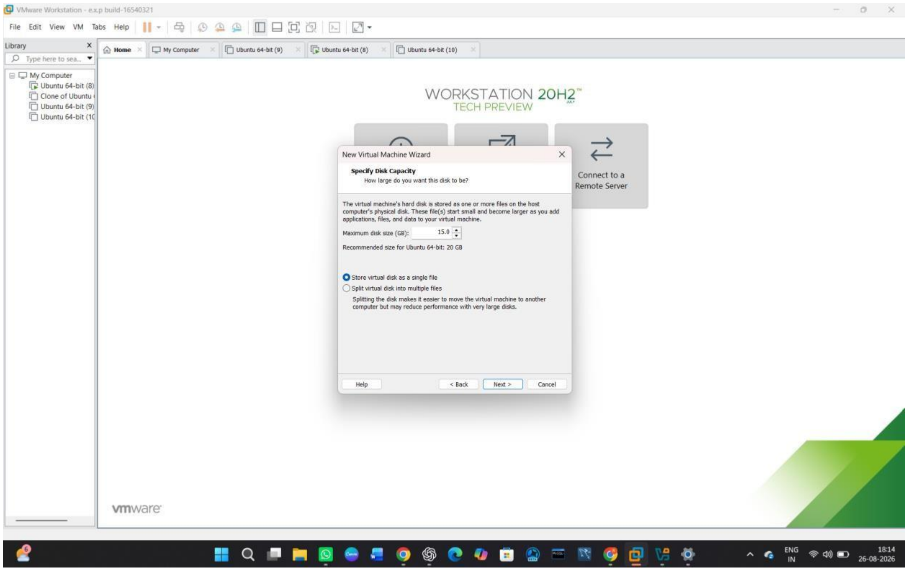
STEP 3: Select Ubuntu ISO



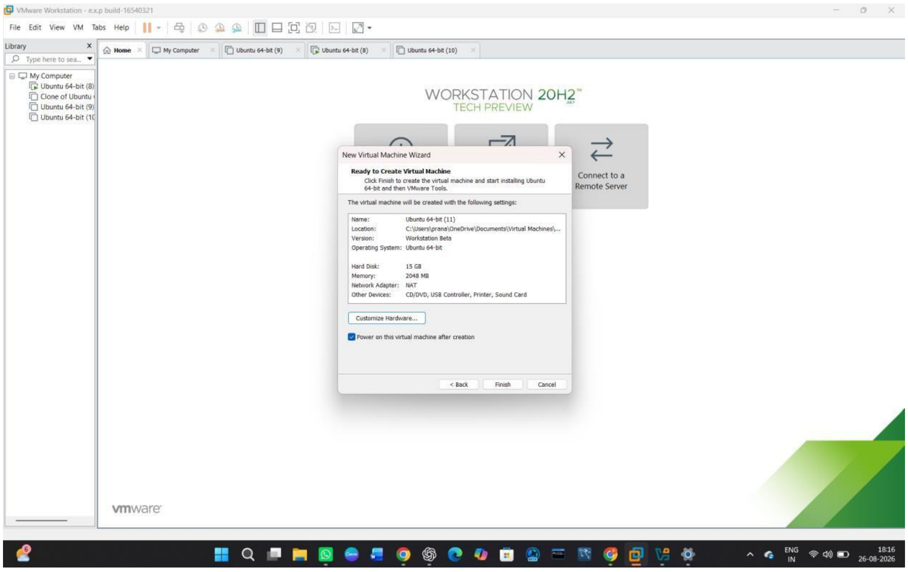
STEP 4: Give the VM a Name



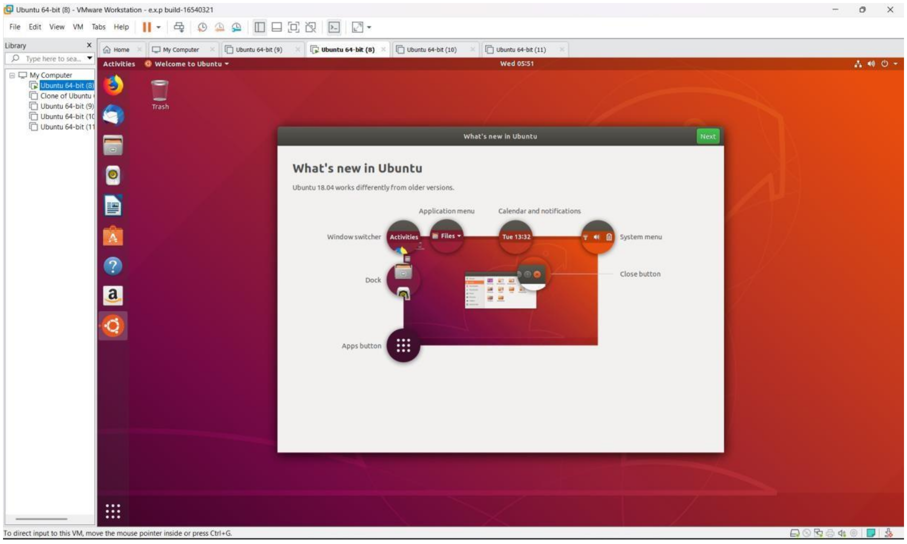
STEP 5: Configure the Virtual Disk



STEP 6: Customize Hardware



STEP 7: Start the Virtual Machine



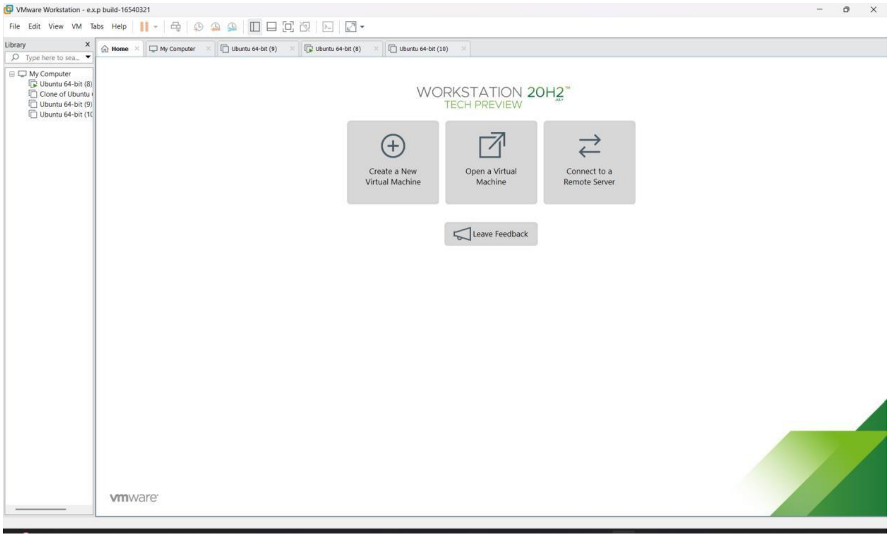
EXP NO 21:Install Virtualbox/VMware Workstation with different flavours of linux or windows OS on top of windows 7 or 8.

AIM:

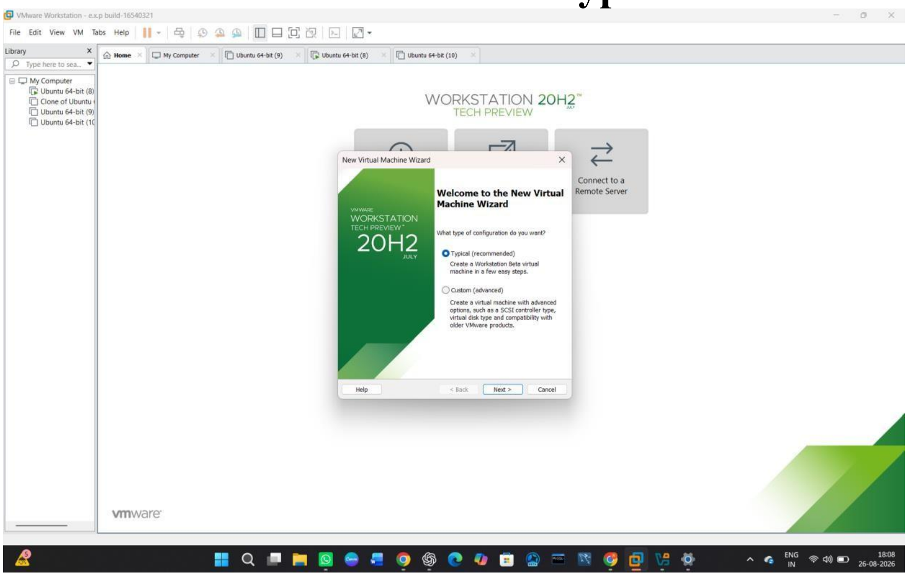
To install VMware Workstation and install a different flavour of Linux/Windows Operating System on top of the existing Windows 7/8 host

PROCEDURE:

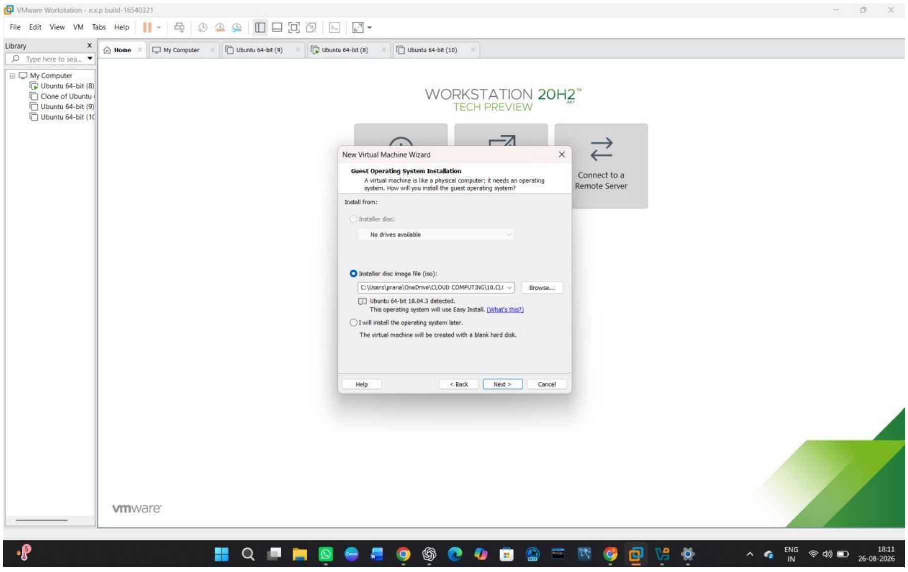
STEP 1: Go to **VMware Workstation.**



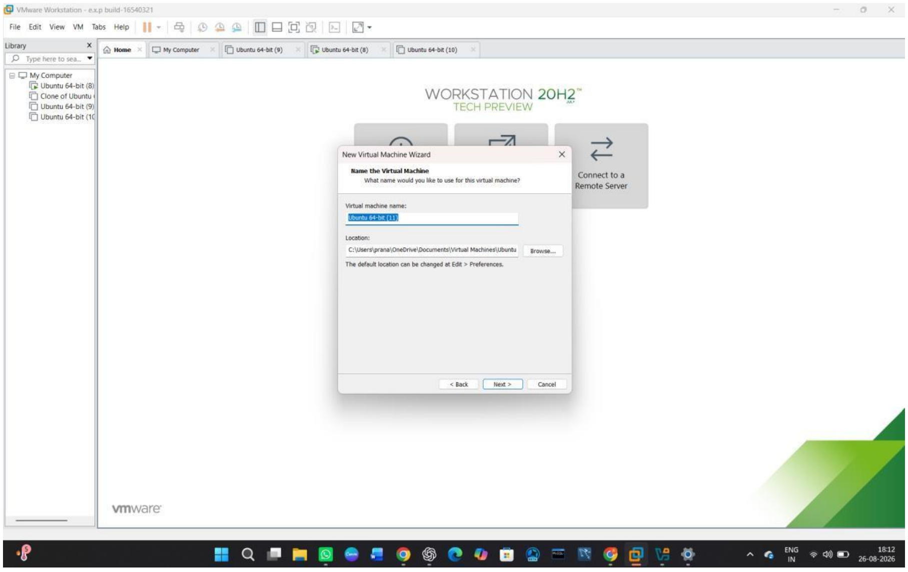
STEP 2 — Select the Virtual Machine Type



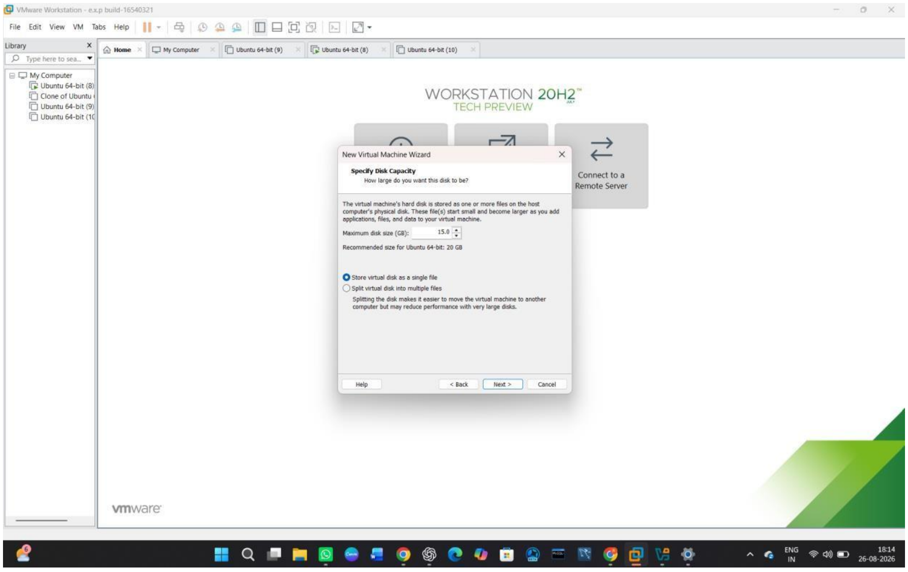
STEP 3 — Select the Ubuntu ISO



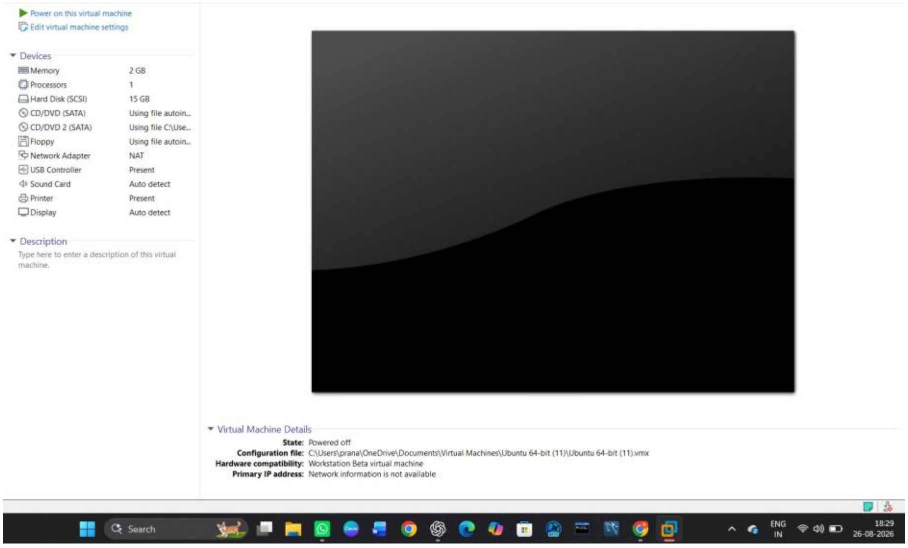
STEP 4 — Name the Virtual Machine



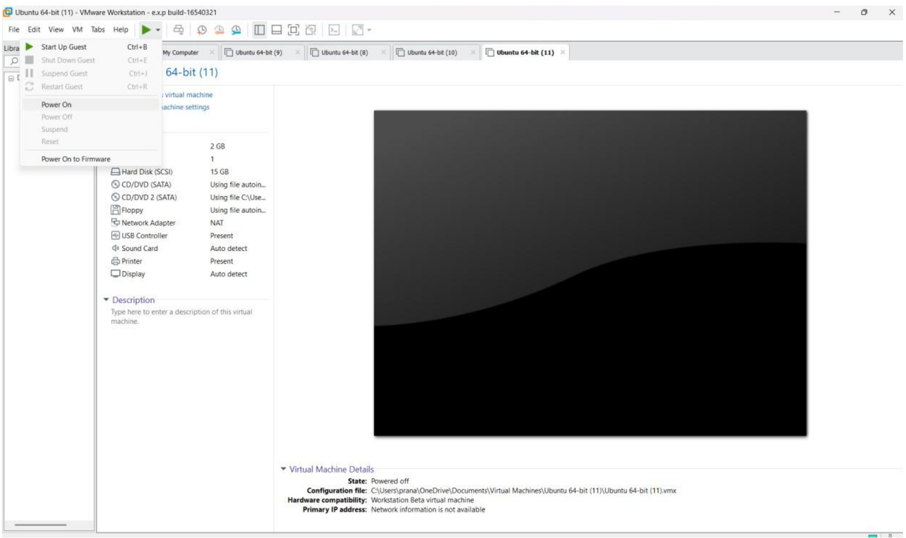
STEP 5 — Configure Disk Capacity



STEP 6 — Customize Hardware

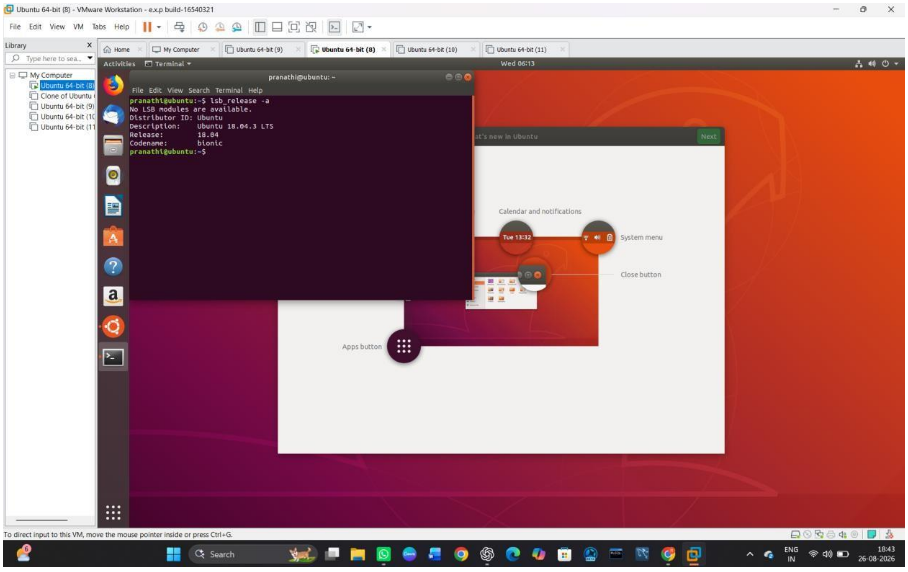


STEP 7 — Power On the Virtual Machine



STEP 8 — Install Ubuntu and Login

STEP 9 — Verify the Installation



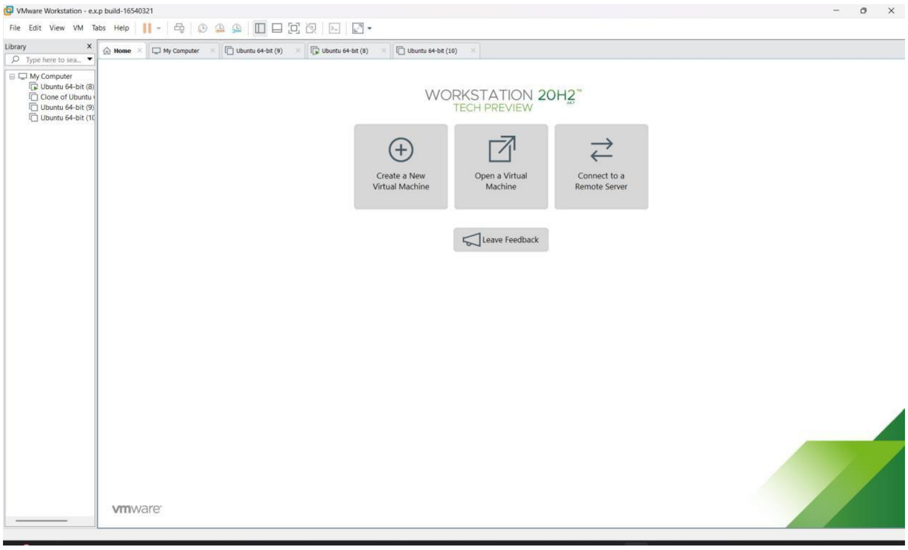
EXP NO 22:To write a procedure to run the virtual machine of different configuration and to check how many virtual machines can be utilized at a particular time.

AIM:

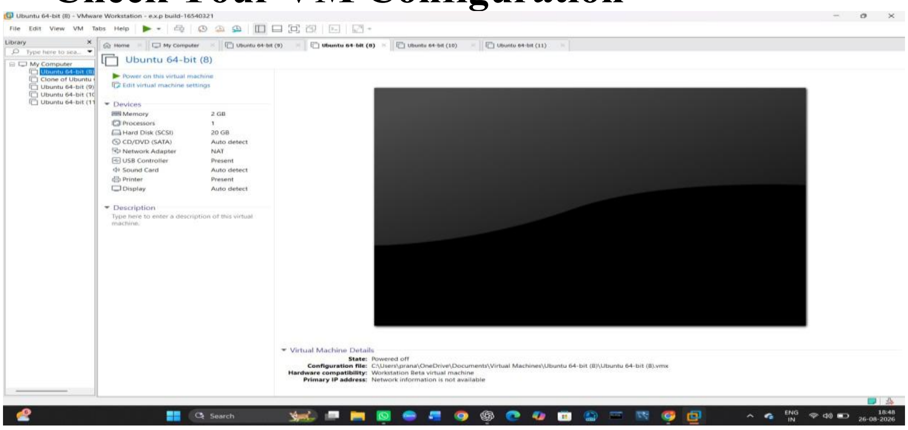
To run virtual machines with different configurations and determine how many virtual machines can be utilized simultaneously on a host system.

PROCEDURE:

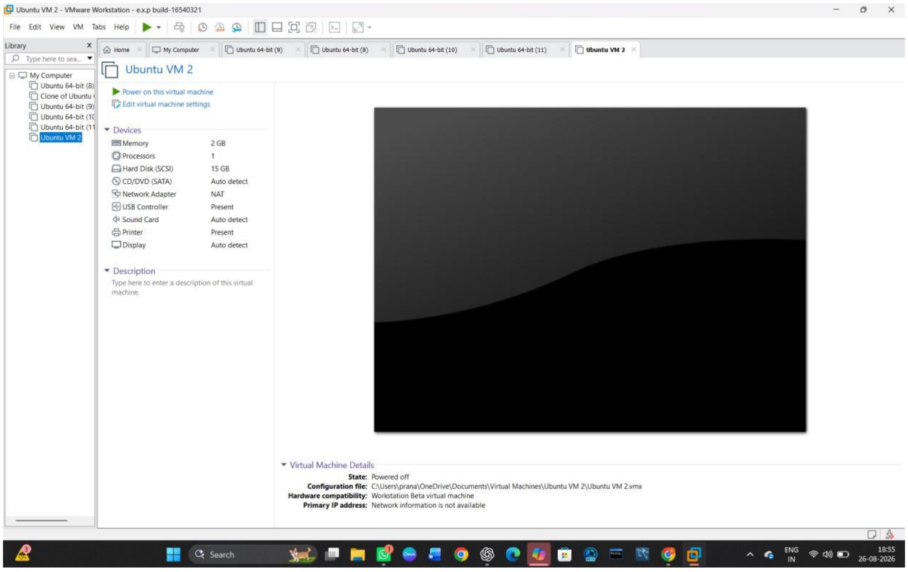
STEP 1— Go to **VMware Workstation**.



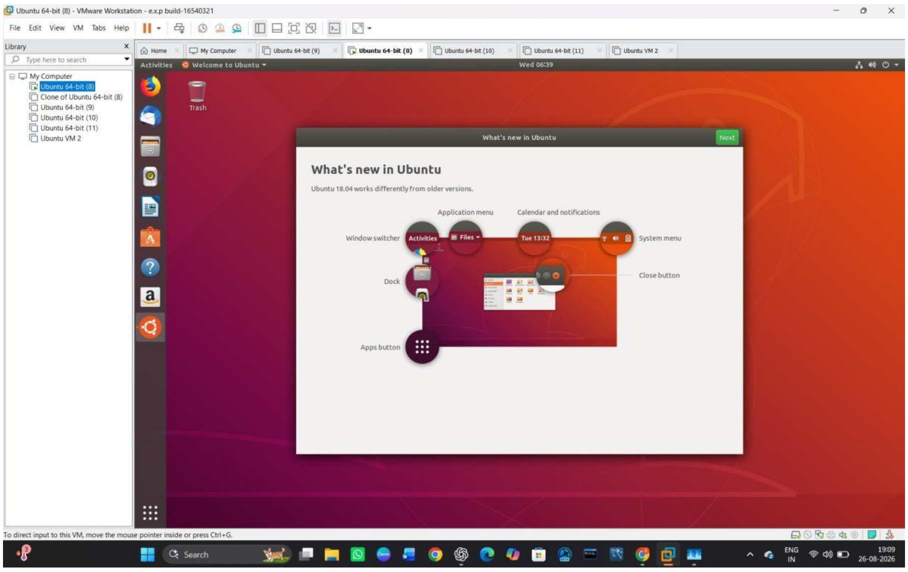
STEP 2 — Check Your VM Configuration



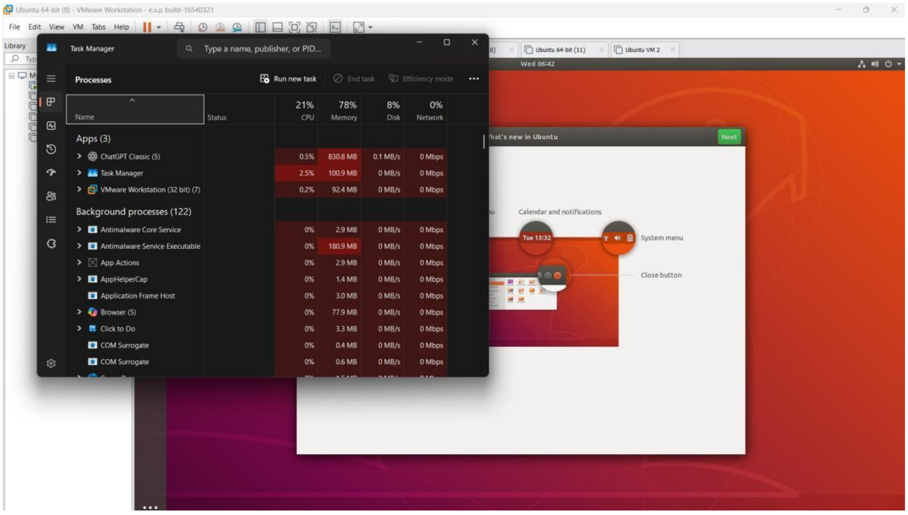
STEP 3 — Create the Second Virtual Machine



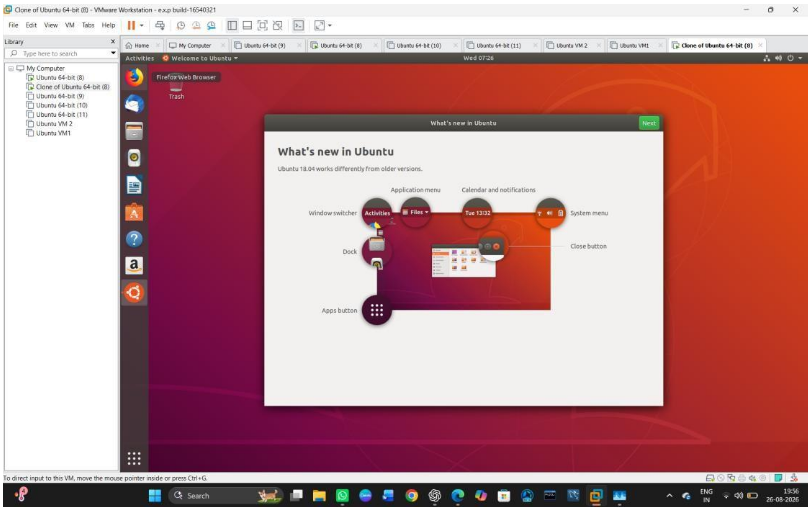
STEP 4 — Start the First VM



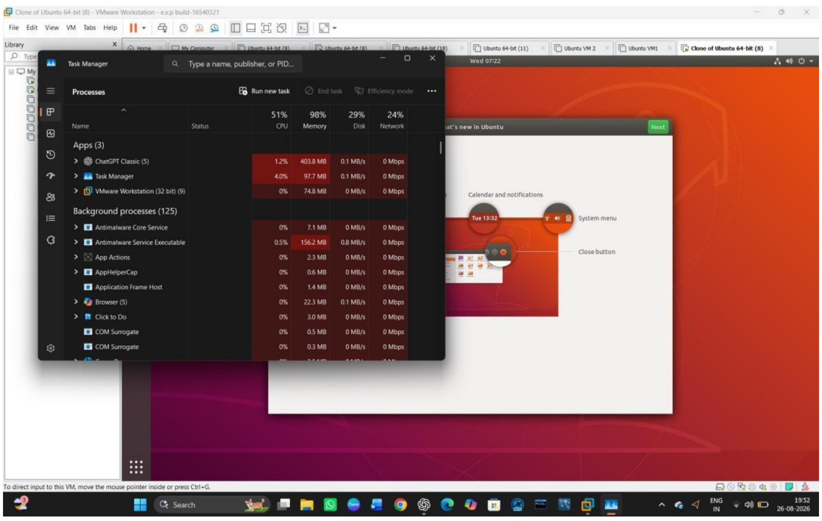
STEP 5 — Check the Host Resources



STEP 6 — Start the Second VM



STEP 7 — Check the Second VM



Result

Two virtual machines were successfully run simultaneously using VMware Workstation. The CPU and memory utilization were monitored using Windows Task Manager. When two VMs were running, memory utilization reached 98%, showing that the available system resources limited the number of VMs that could be utilized simultaneously.

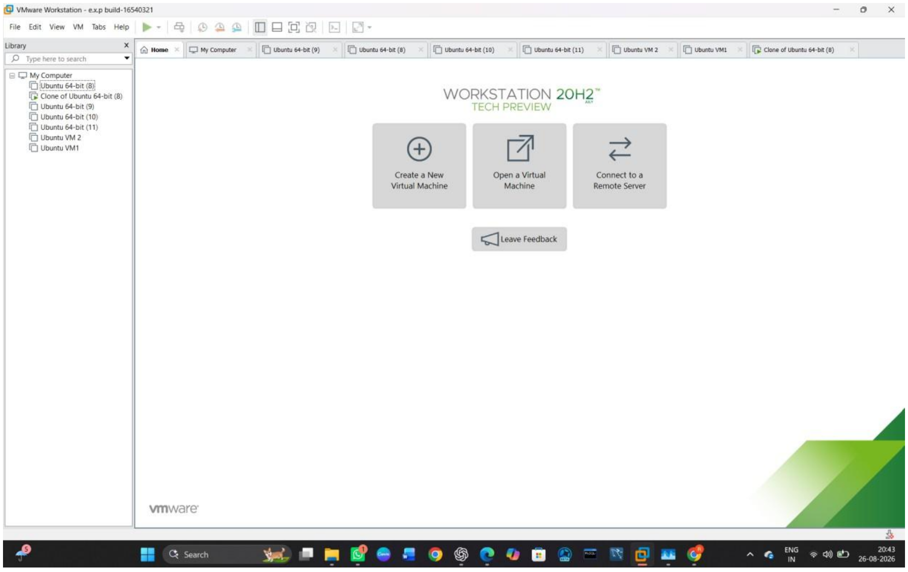
EXP NO 23:Write a Procedure To Attach Virtual Block To The Virtual Machine And Check Whether It Holds the Data Even After The Release Of The Virtual Machine

AIM:

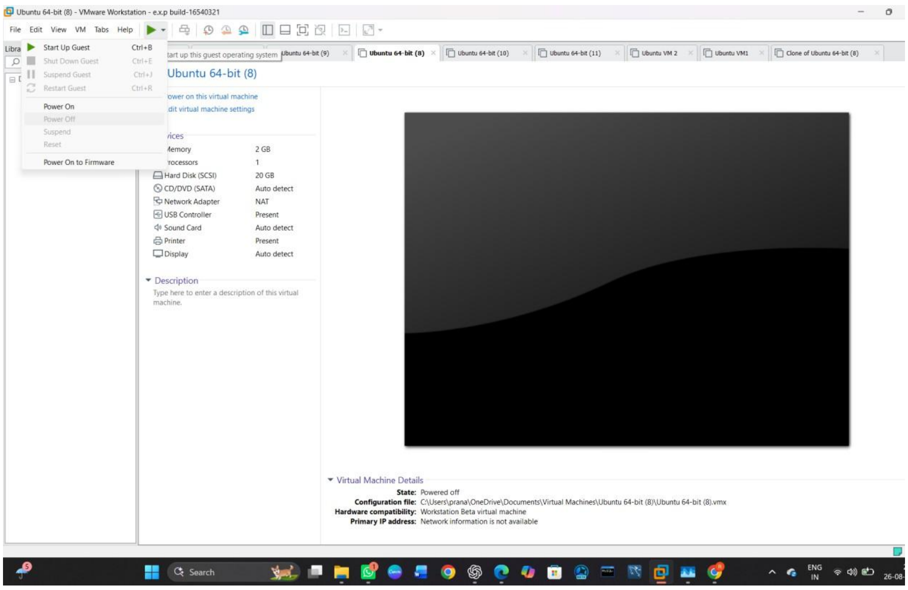
To attach a virtual block storage disk to a virtual machine, store data in it, release/detach the virtual machine, and verify that the data is retained.

PROCEDURE:

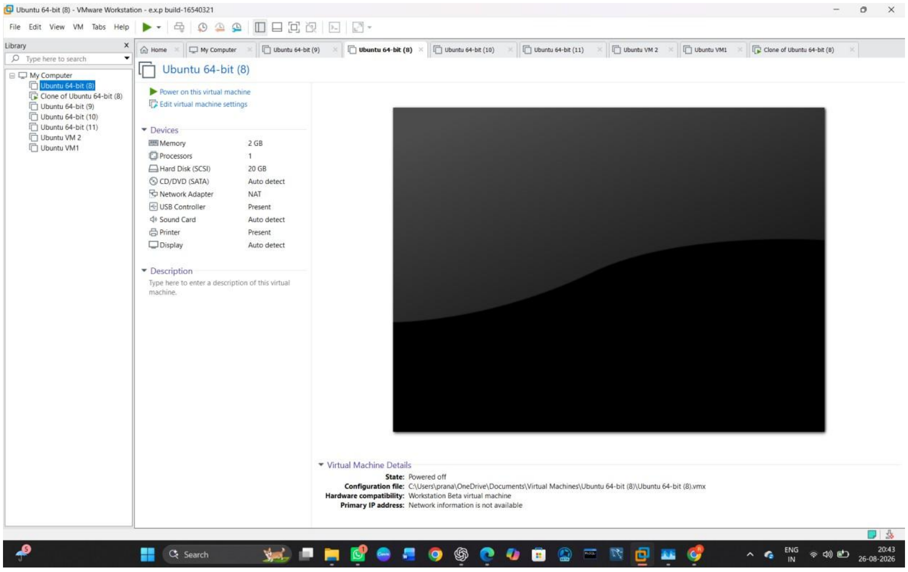
STEP 1: Open **VMware Workstation**.



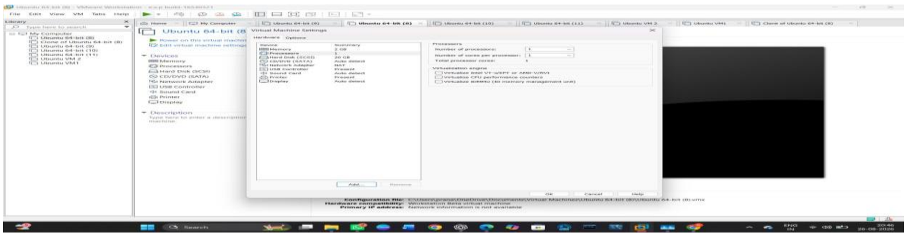
STEP 2: Select the Ubuntu Virtual Machine and Power Off Ubuntu



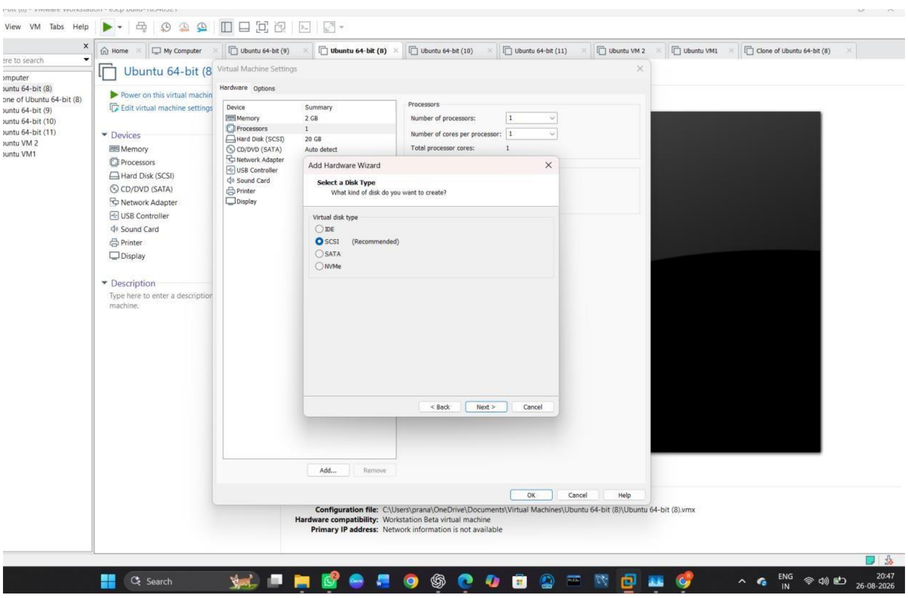
STEP 3 — Open Virtual Machine Settings



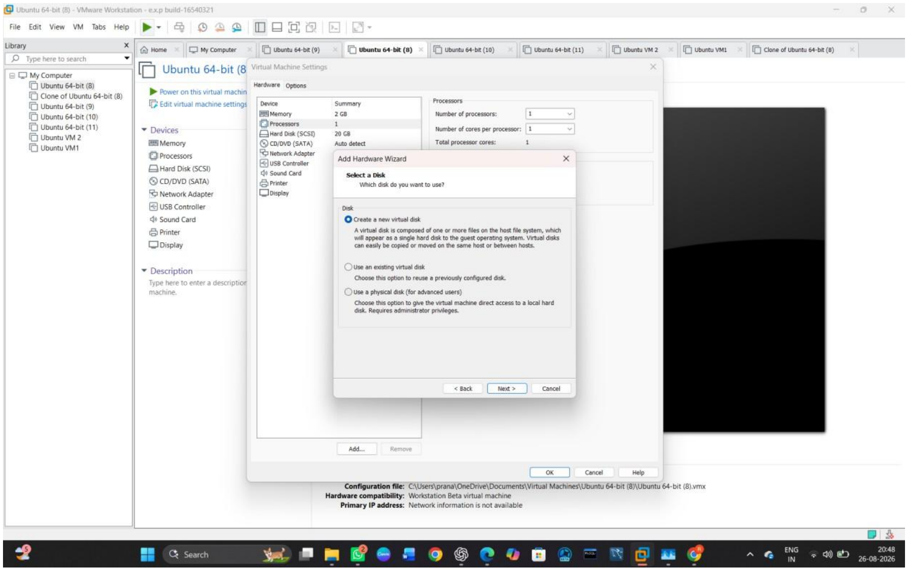
STEP 4 — Add a New Virtual Hard Disk



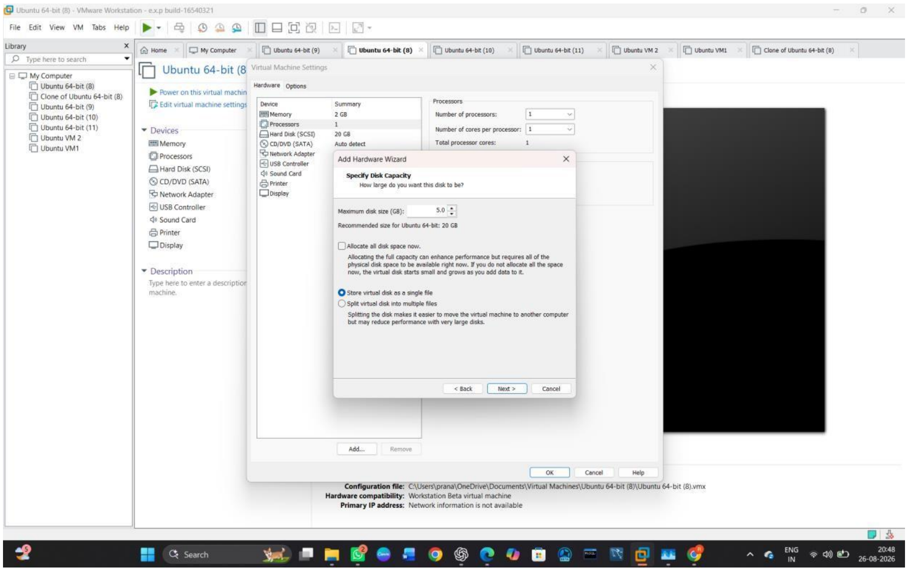
STEP 5 — Select Disk Type



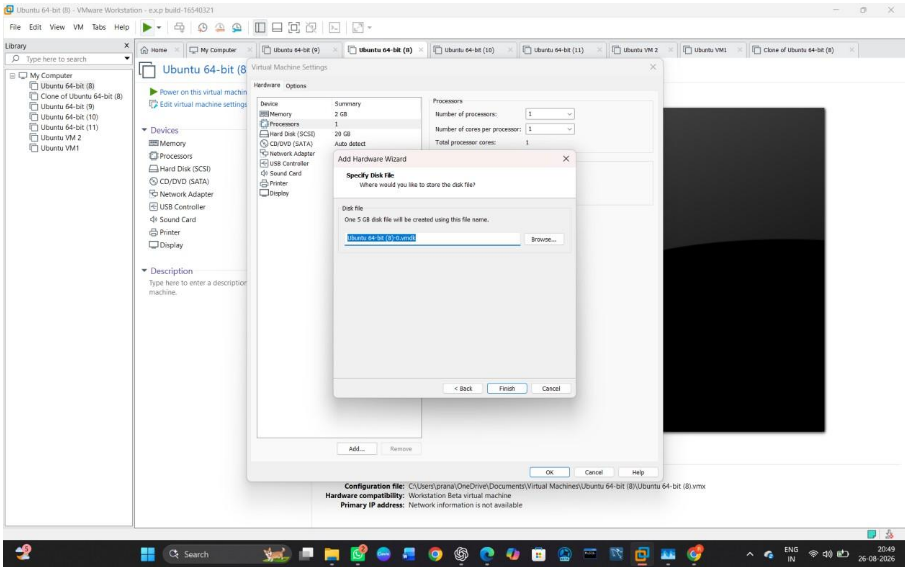
STEP 6 — Create a New Virtual Disk



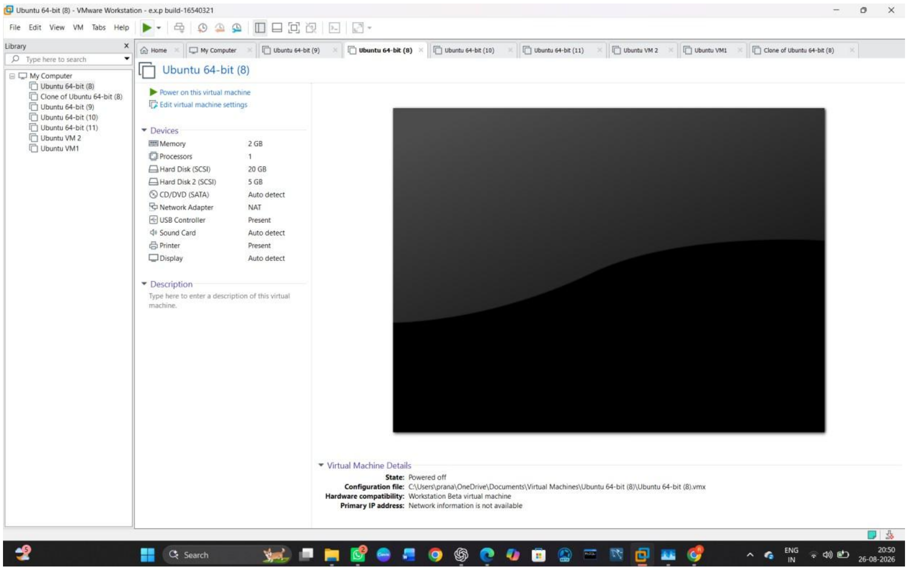
STEP 7— Set Disk Capacity



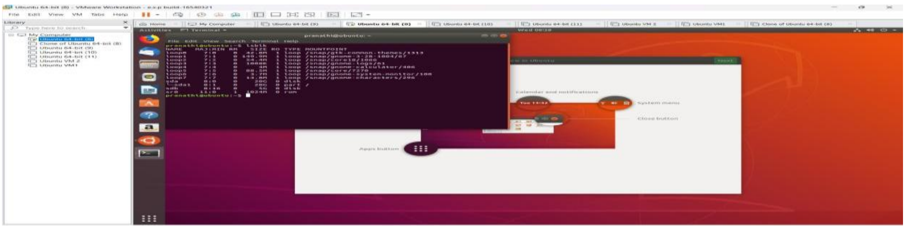
STEP 8 — Specify the Disk File



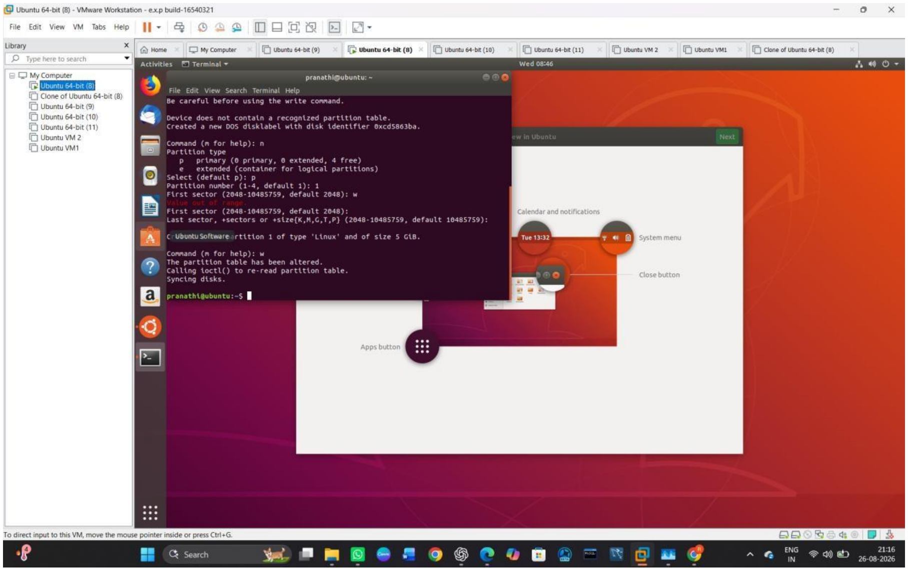
STEP 9 — Confirm the New Disk



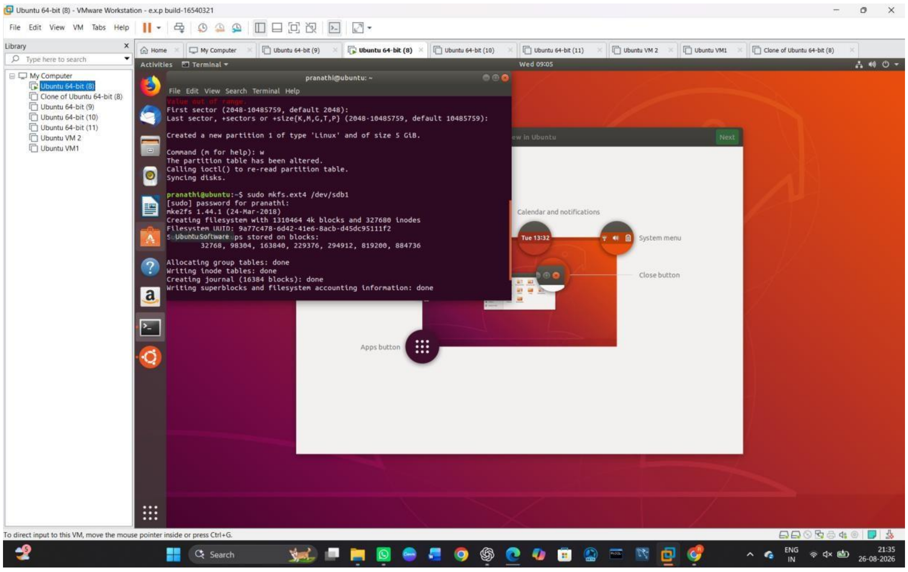
STEP 10 — Check the New Virtual Disk



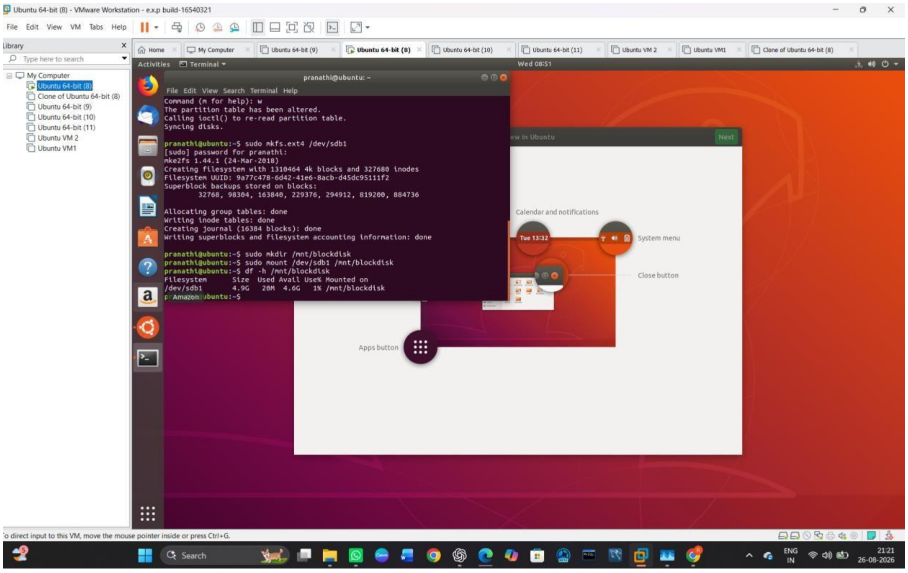
STEP 11 — Create a Partition on the New Disk



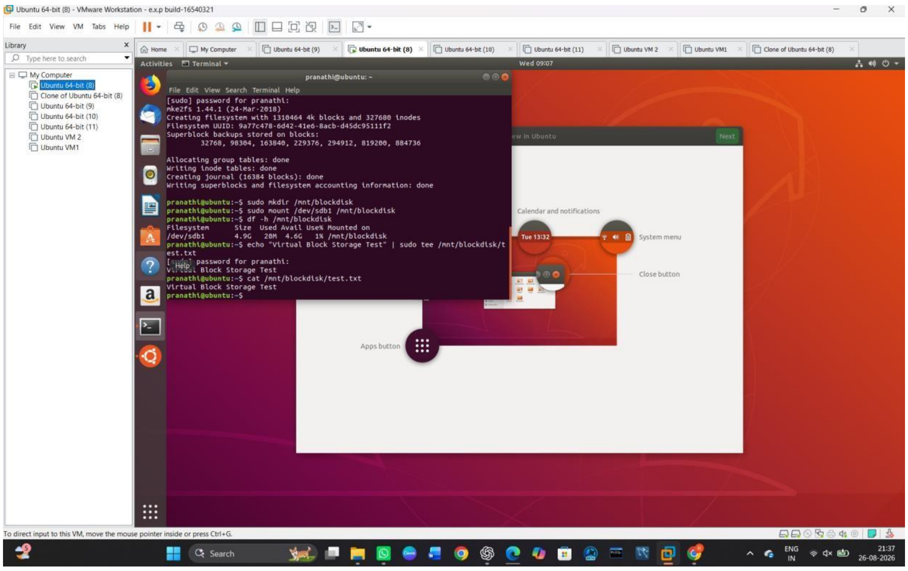
STEP 12 — Format the New Partition



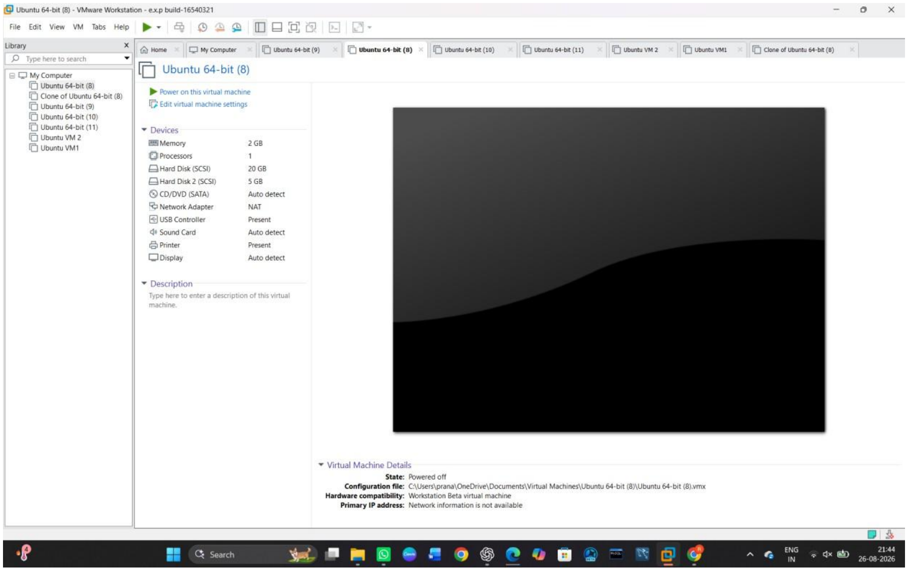
STEP 13 — Create a Mount Point



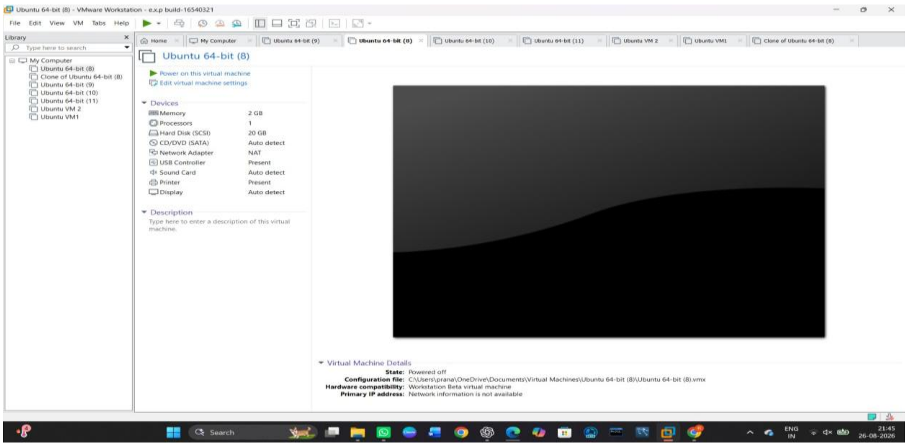
STEP 14 — Create Test Data



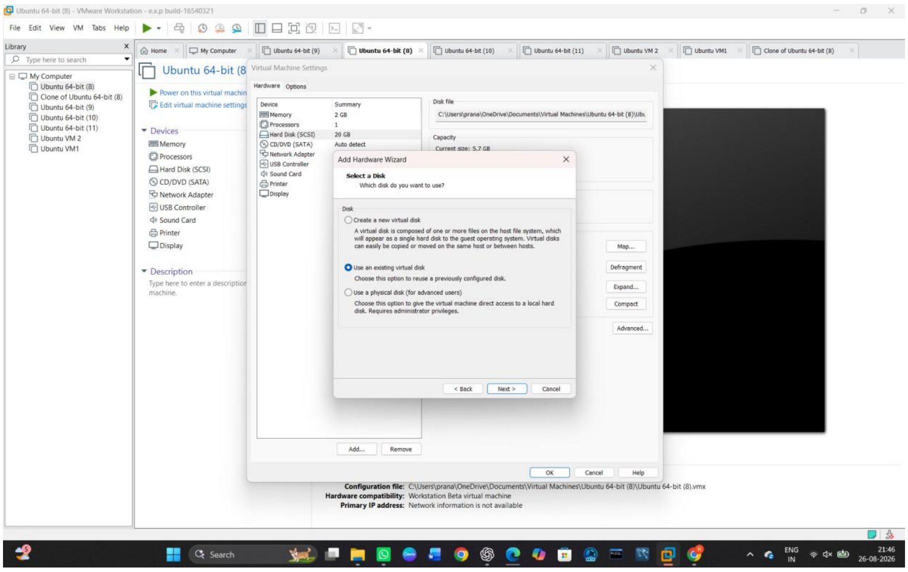
STEP 15 — Shut Down the Virtual Machine



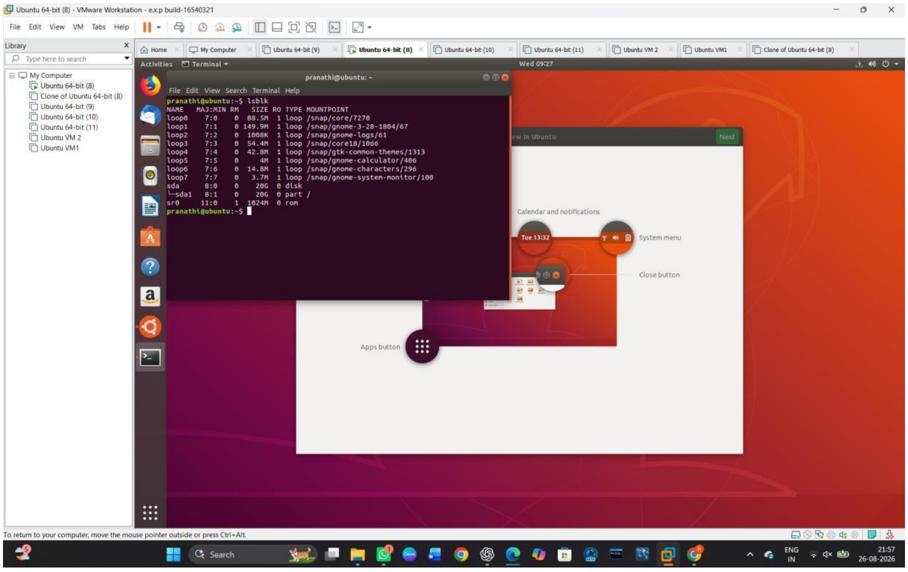
STEP 16 — Detach the Virtual Disk



STEP 17 — Attach the Existing Virtual Disk



STEP 18 — Start Ubuntu Again



Result

The virtual block disk was successfully attached to the virtual machine. Data was stored on the disk, the disk was detached and reattached, and the previously stored data was successfully recovered. Hence, the virtual block storage retains data even after the release of the virtual machine.

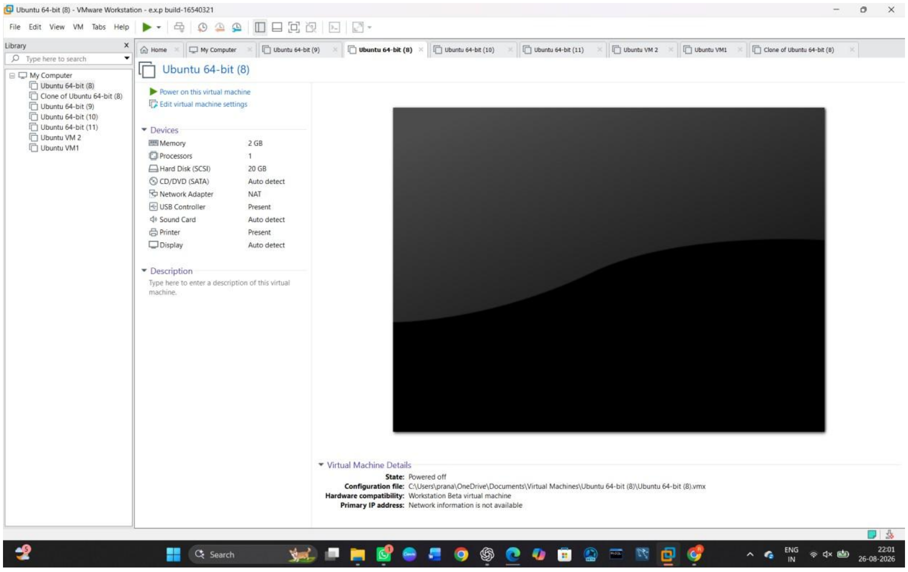
EXP NO24:To showcase the virtual machine migration based on the certain condition from one host to the other.

AIM:

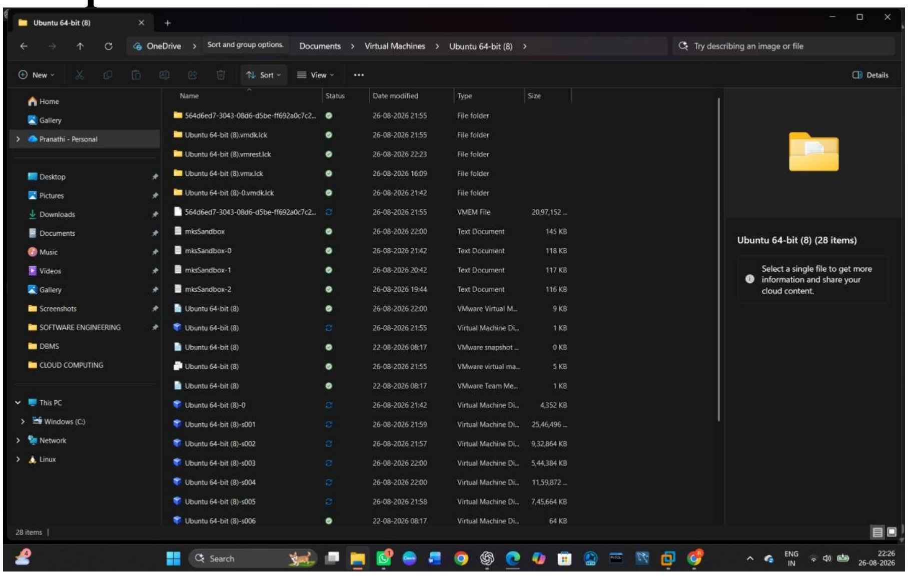
To showcase virtual machine migration from one host to another based on a certain condition.

Procedure:

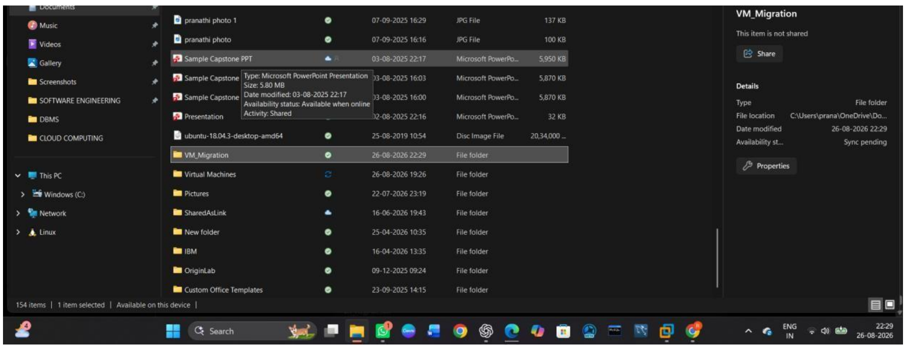
STEP 1 — Prepare the Source VM



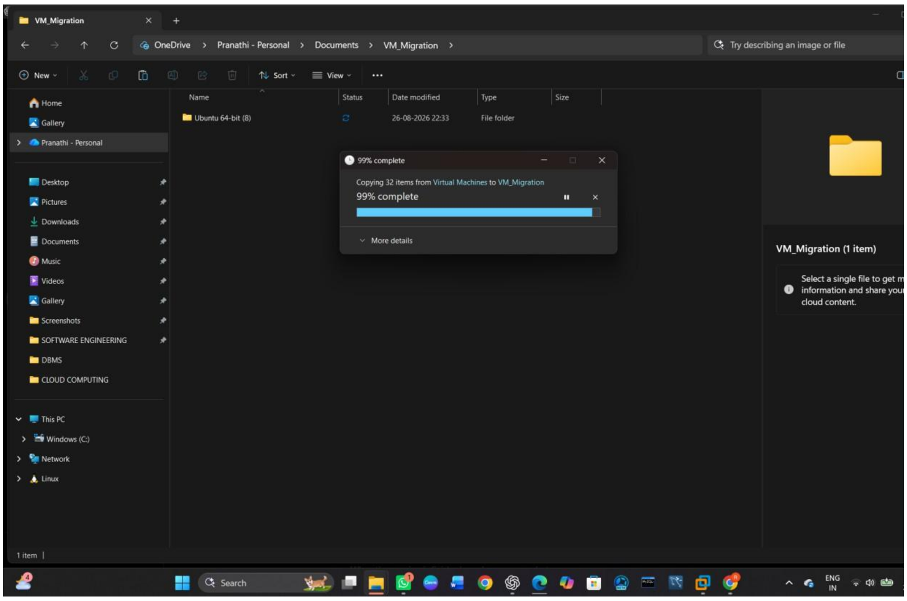
STEP 2 — Open the VM Folder



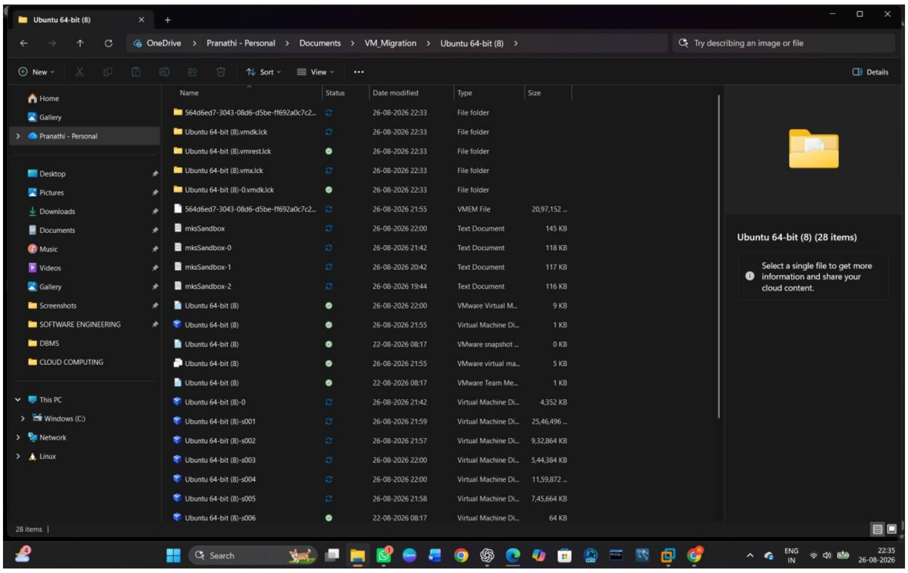
STEP 3 — Create the Destination Folder



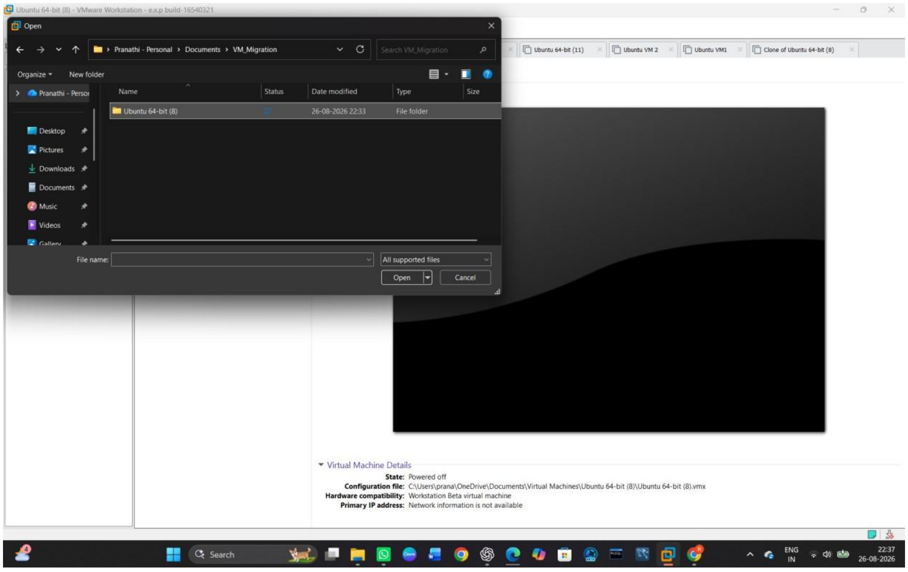
STEP 4 — Copy the Ubuntu VM



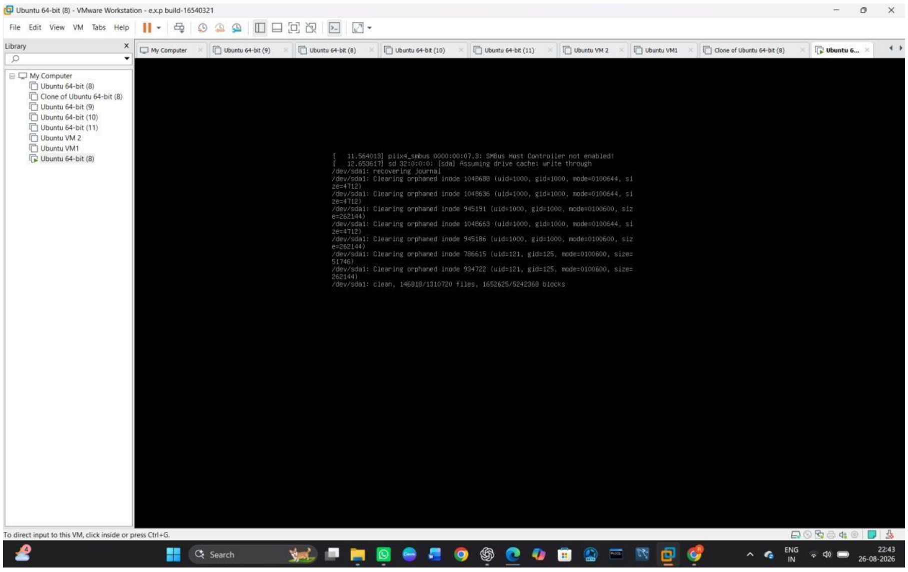
STEP 5 — Open the Copied VM Folder



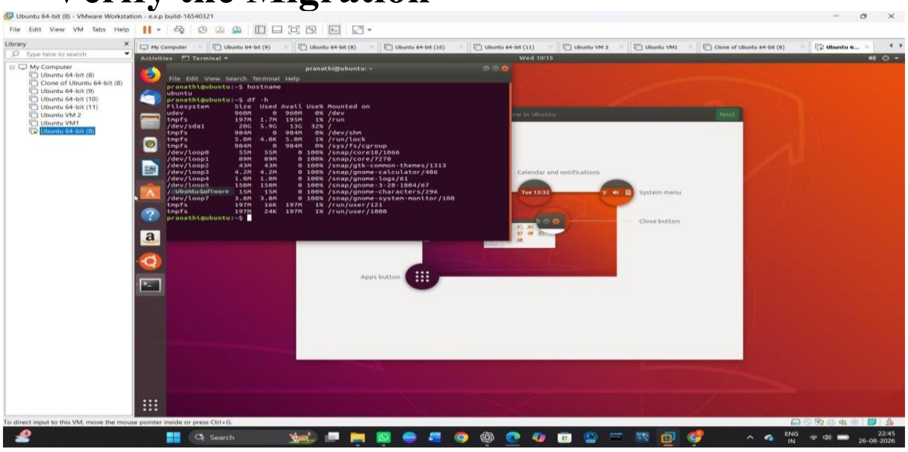
STEP 6 — Load the migrated VM in VMware Workstation.



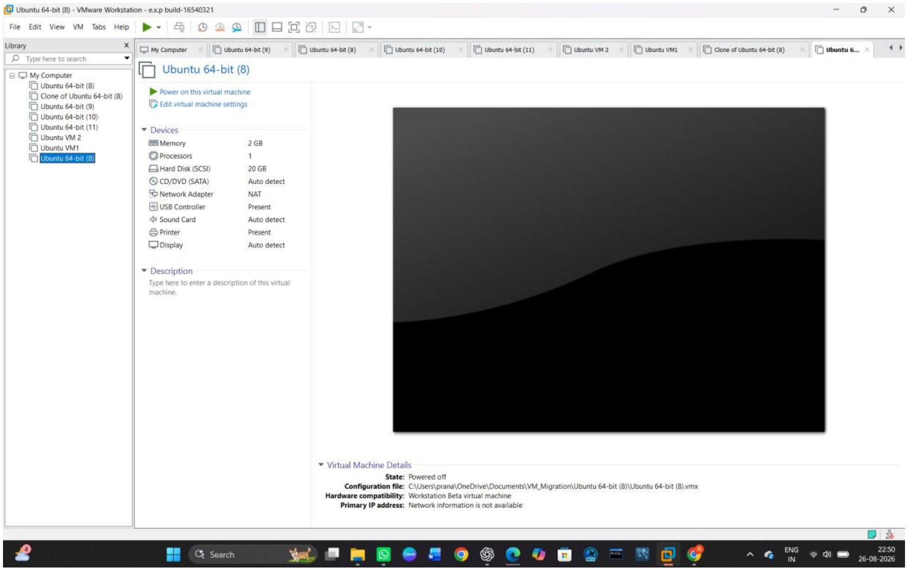
STEP 7 — Start the Migrated VM



STEP 8 — Verify the Migration



STEP 9 — Shut Down the Migrated VM



STEP 10 — Final Verification